Simulated Annealing

Simulated Annealing Steps

1. Select a start node (root node).
2. Randomly select a child of the current node, calculate a value reflecting how good such child is like $\boldsymbol{value(node) = -heuristic(node)}$.
3. Select the child if it is better than the current node. Else try another child.

   A node is better than the current node if $\boldsymbol{\Delta E = value[next] - value[current] > 0}$.

   Else if $\boldsymbol{\Delta E < 0}$, then try to find another child.

4. If the child was not better than the current node then it will be selected with probability equal to $\mathbf{p} = \boldsymbol{e^{\frac{\Delta E}{T}}}$

   where $\boldsymbol{\Delta E = value[next] - value[current]}$

   $\boldsymbol{T}$ is a temperature.

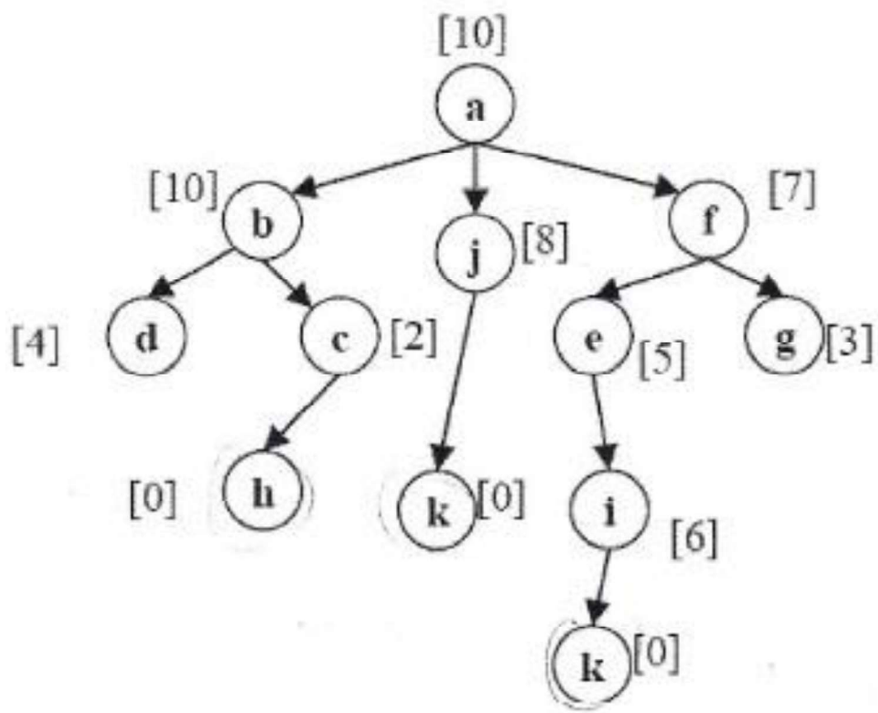
5. Stop if no improvement can be found or after a fixed time.

Simulated Annealing Example – T=10

Current

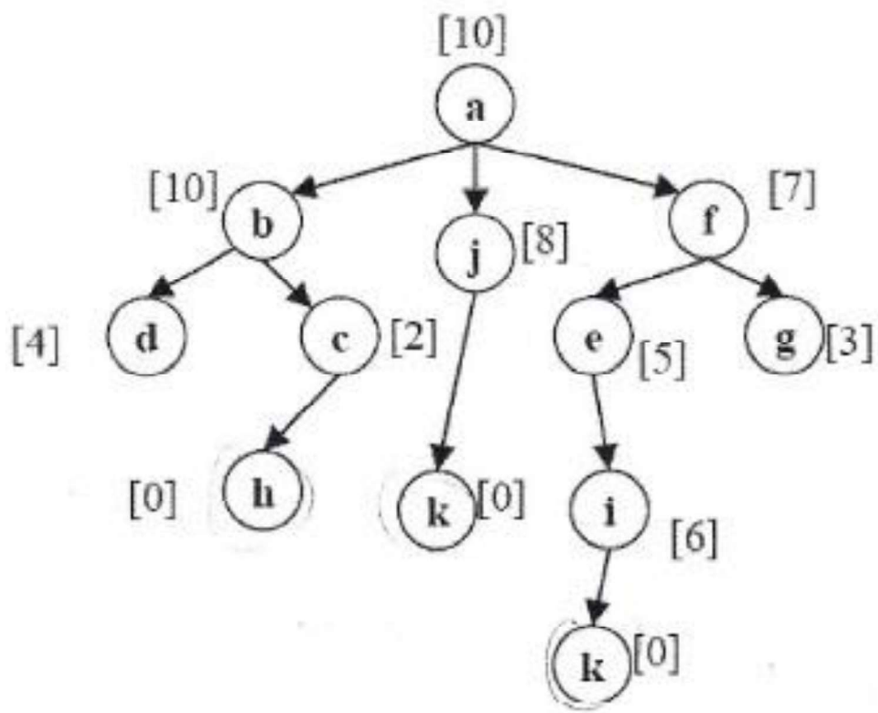
a

Children

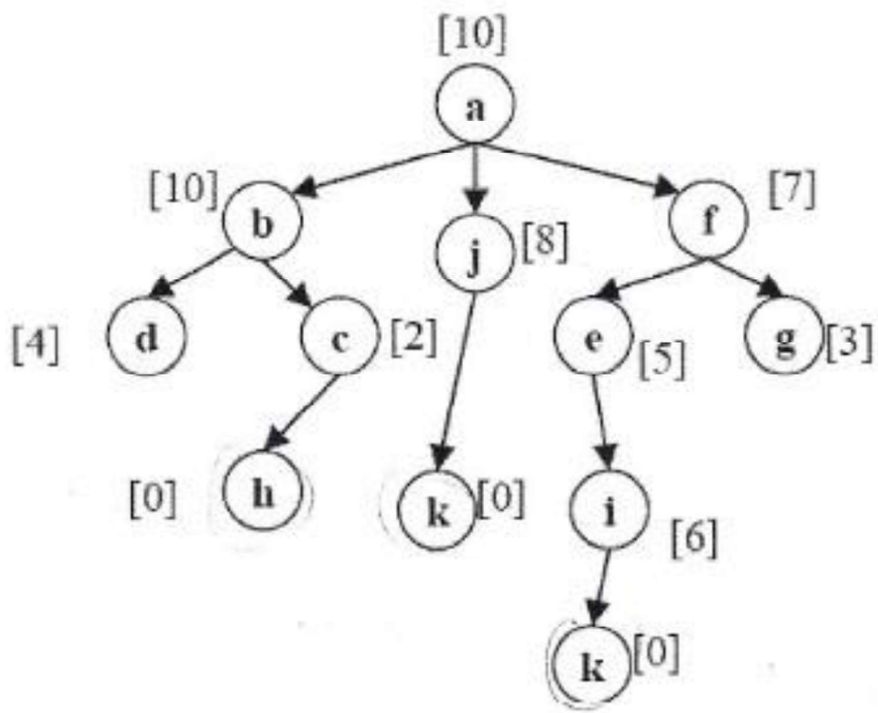


Simulated Annealing Starting from Node a

Current	Children
a	



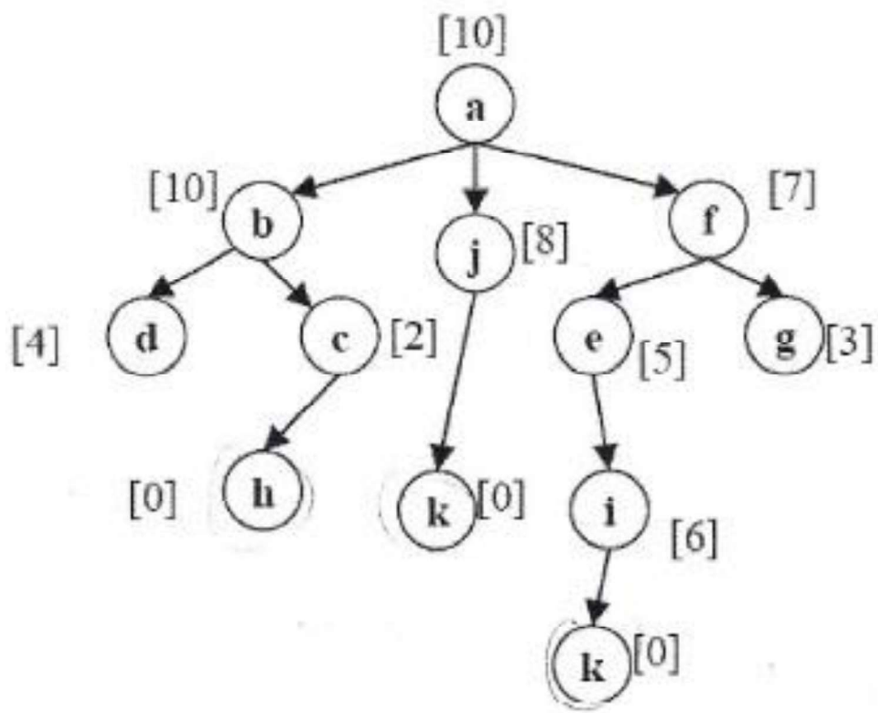
Simulated Annealing Starting from Node a



Current	Children
a	

Simulated Annealing

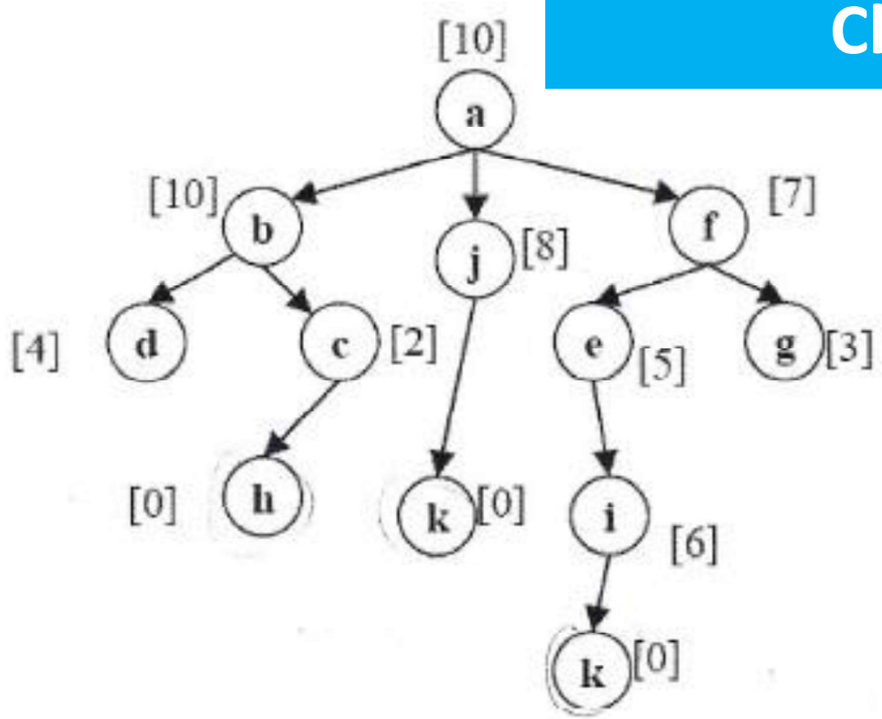
Current	Children
a	---
a	f_7, j_8, b_{10}



Simulated Annealing

Current	Children
a	---
a	f_7, j_8, b_{10}

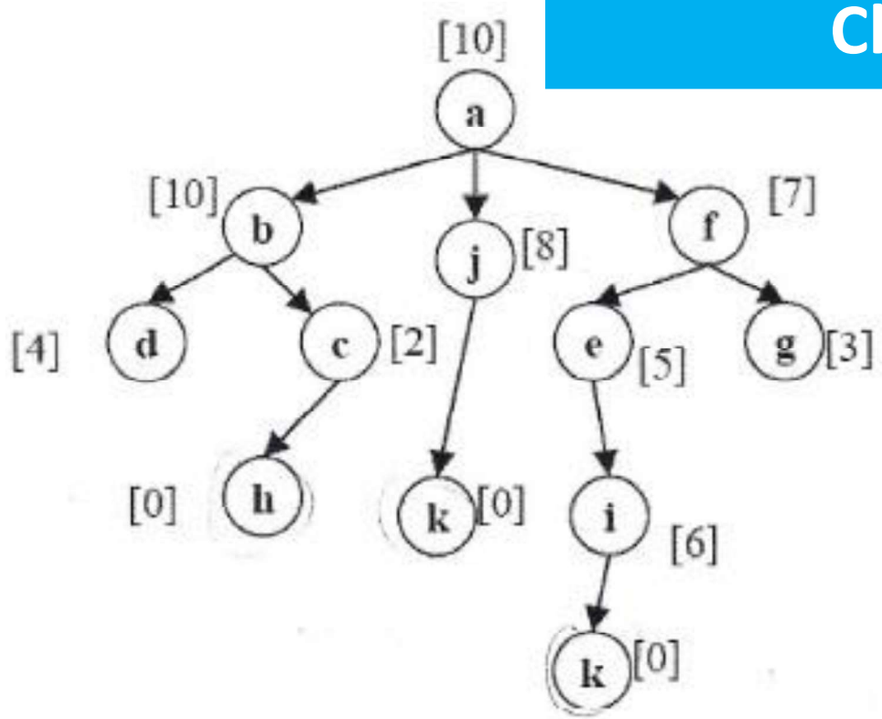
Randomly Select a Child



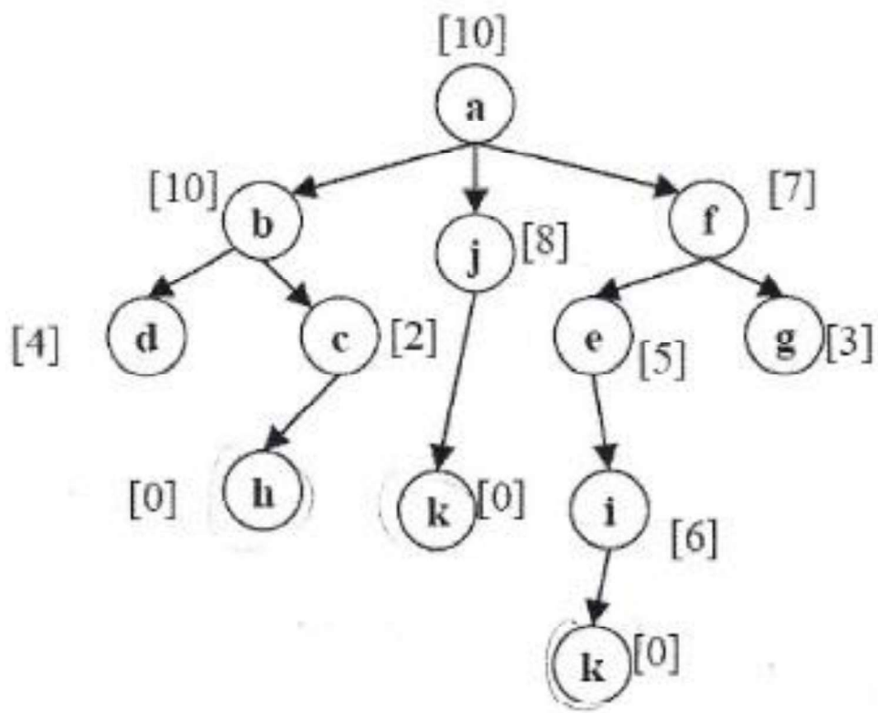
Simulated Annealing

Current	Children
a	---
a	f_7, j_8, b_{10}

Randomly Select a Child



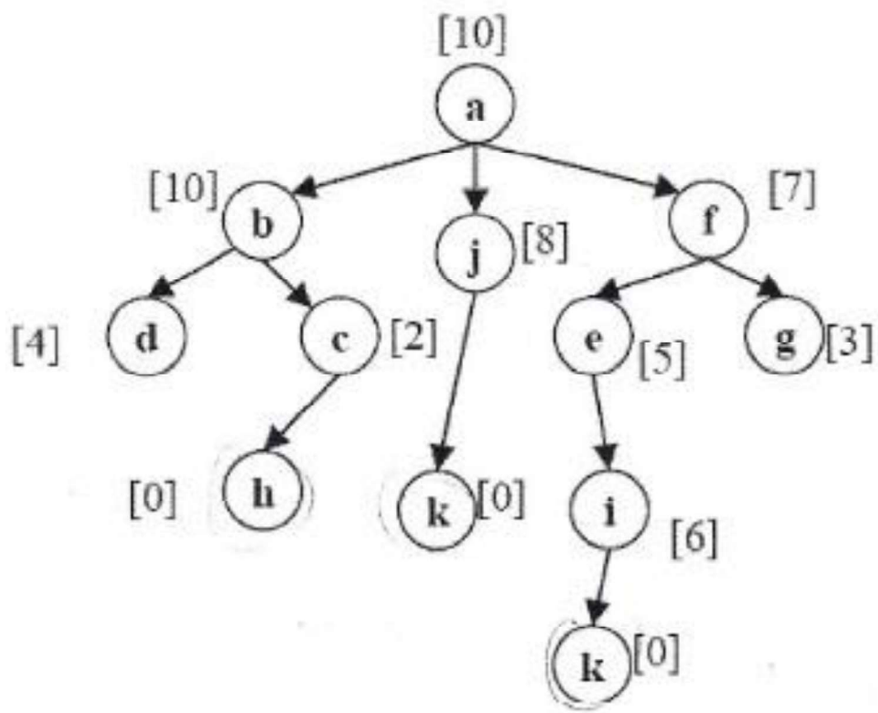
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}

Check if next node f_7 is better than current node

Simulated Annealing

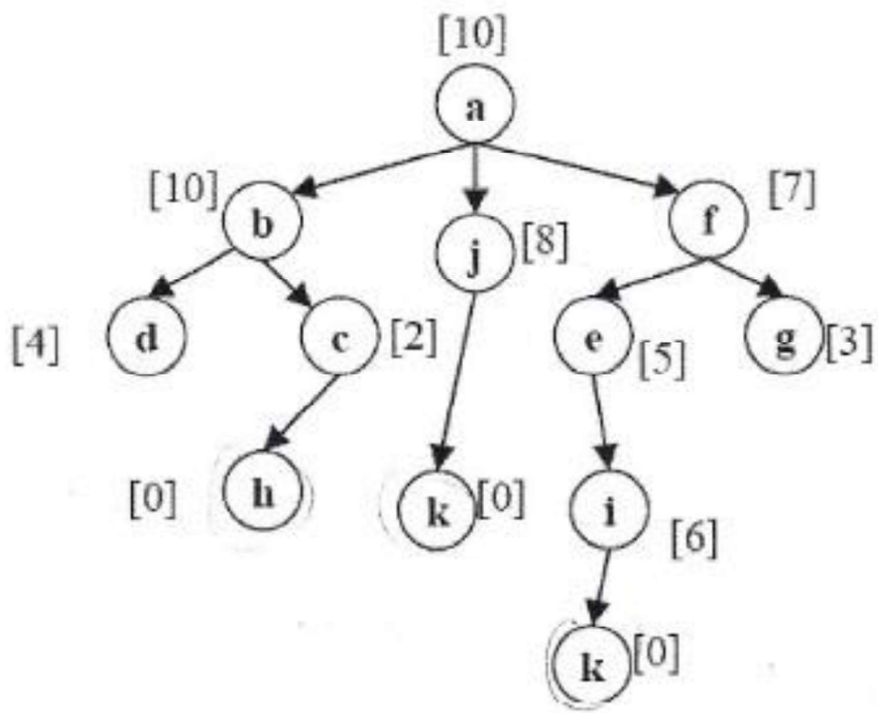


Current	Children
a	---
a	f_7, j_8, b_{10}

Check if next node f_7 is better than current node

$$\Delta E > 0$$

Simulated Annealing

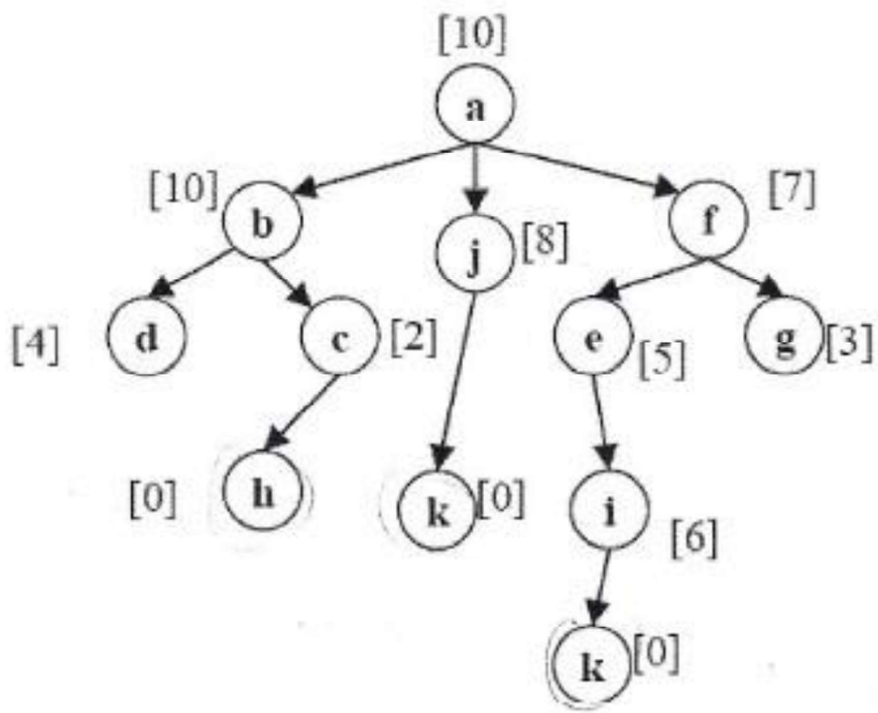


Current	Children
a	---
a	f_7, j_8, b_{10}

Check if next node f_7 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

Simulated Annealing



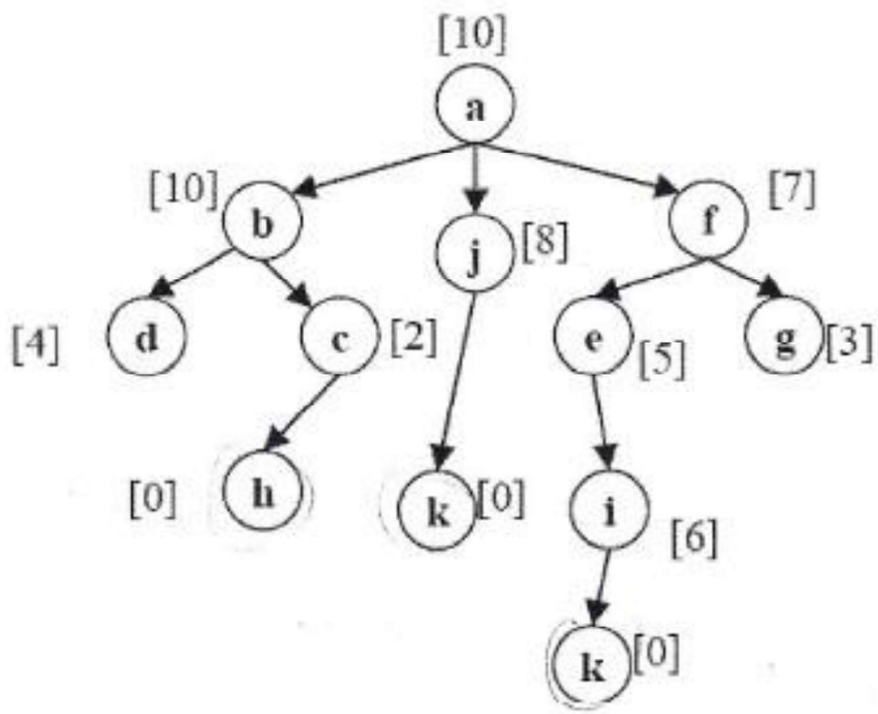
Current	Children
a	---
a	f_7, j_8, b_{10}

Check if next node f_7 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(f_7) - \text{value}(a_{10})$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}

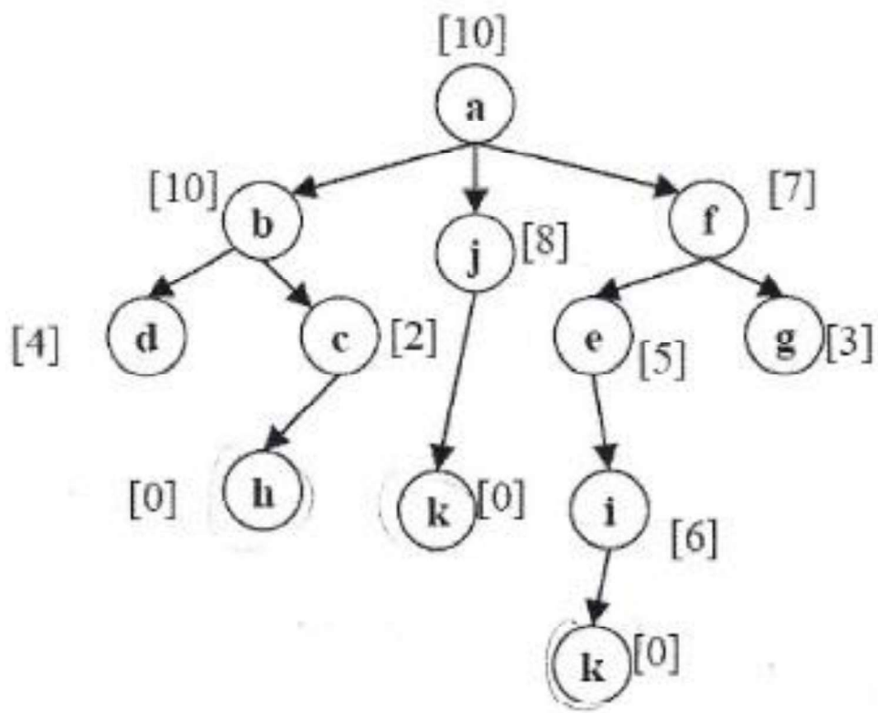
Check if next node f_7 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(f_7) - \text{value}(a_{10})$$

$$\text{value}(f_7) = -\text{heuristic}(f_7) = -7$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}

Check if next node f_7 is better than current node

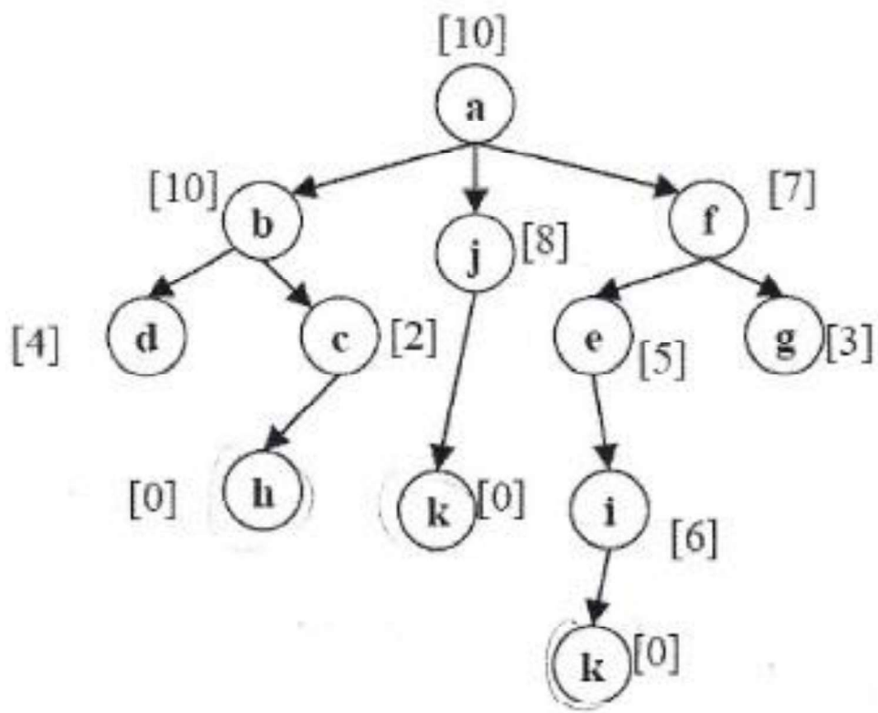
$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(f_7) - \text{value}(a_{10})$$

$$\text{value}(f_7) = -\text{heuristic}(f_7) = -7$$

$$\text{value}(a_{10}) = -\text{heuristic}(a_{10}) = -10$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}

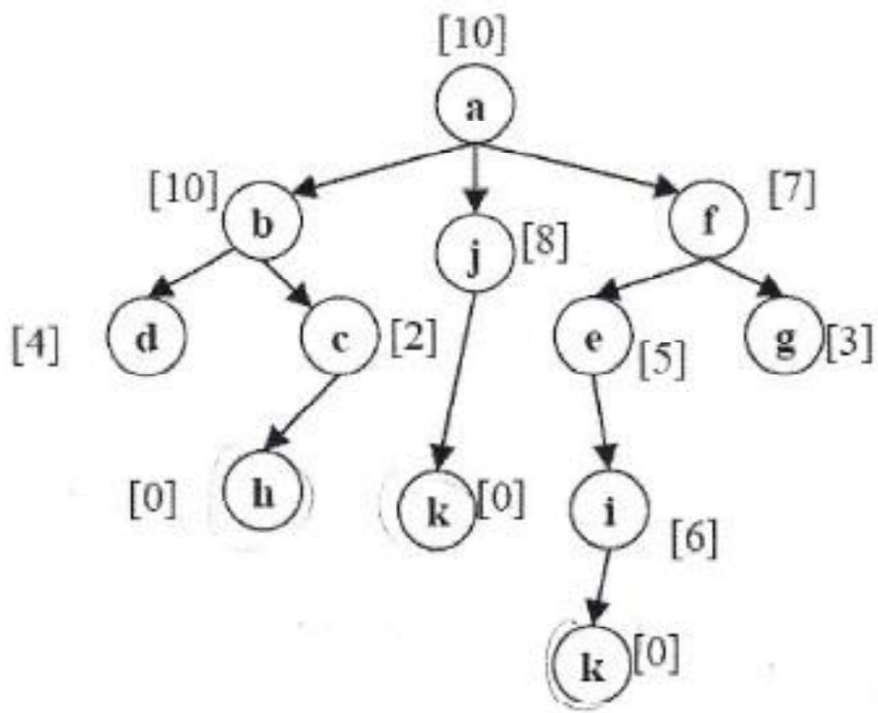
Check if next node f_7 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(f_7) - \text{value}(a_{10})$$

$$\Delta E = -7 - (-10) = +3$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}

Check if next node f_7 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

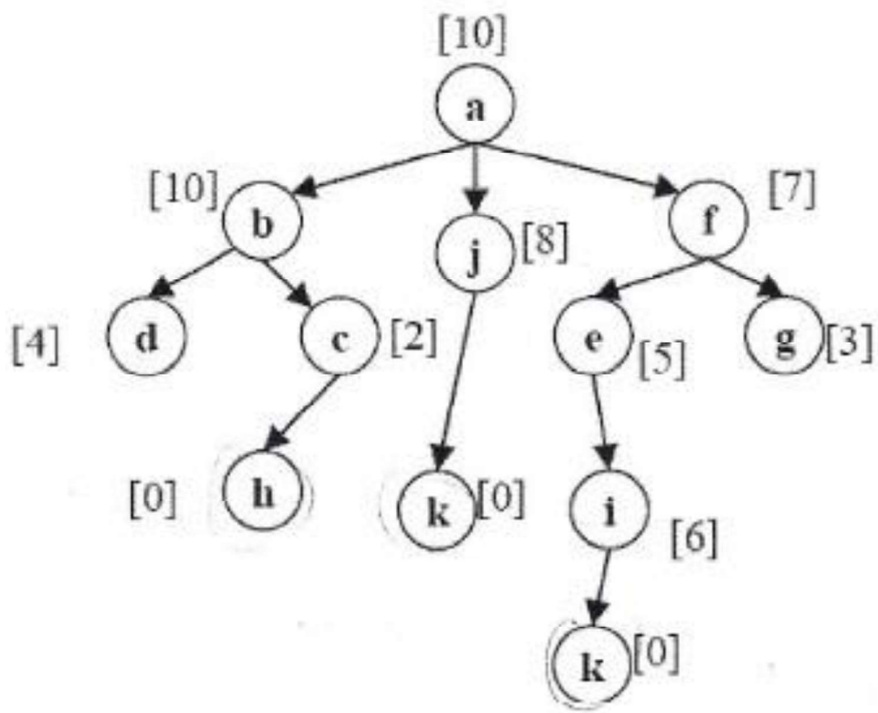
$$\Delta E = \text{value}(f_7) - \text{value}(a_{10})$$

$$\Delta E = -7 - (-10) = +3$$

$$\because \Delta E > 0$$

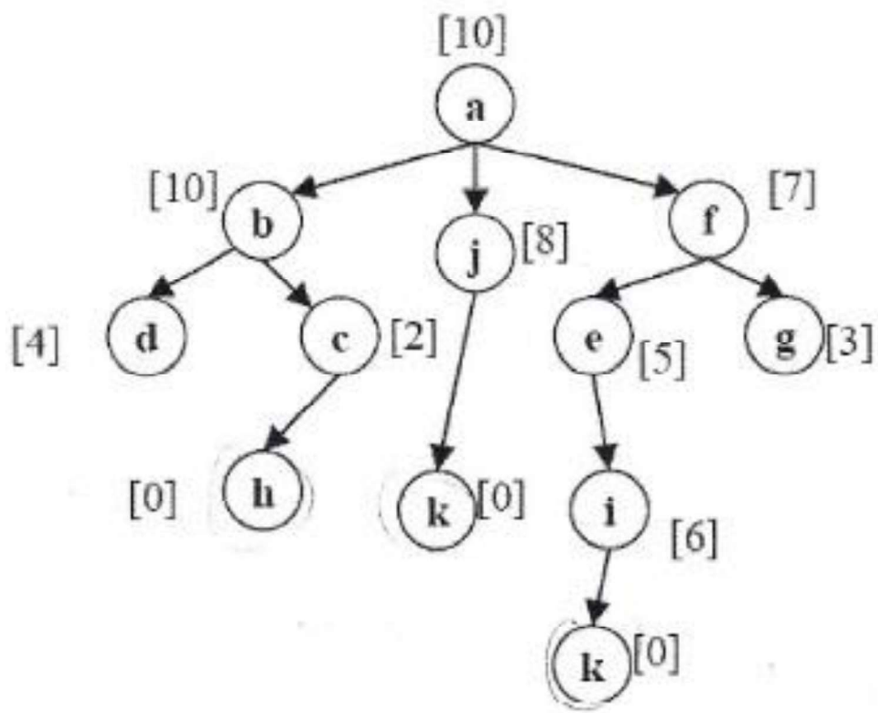
$\therefore f_7$ **will be selected with probability 1**

Simulated Annealing



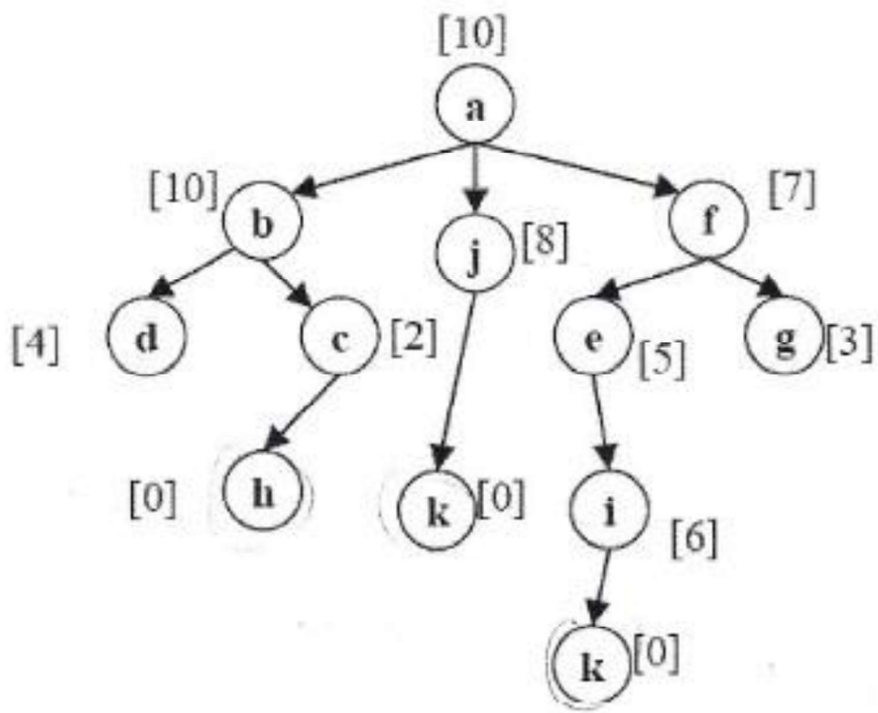
Current	Children
a	---
a	f_7, j_8, b_{10}
f	

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	

Simulated Annealing

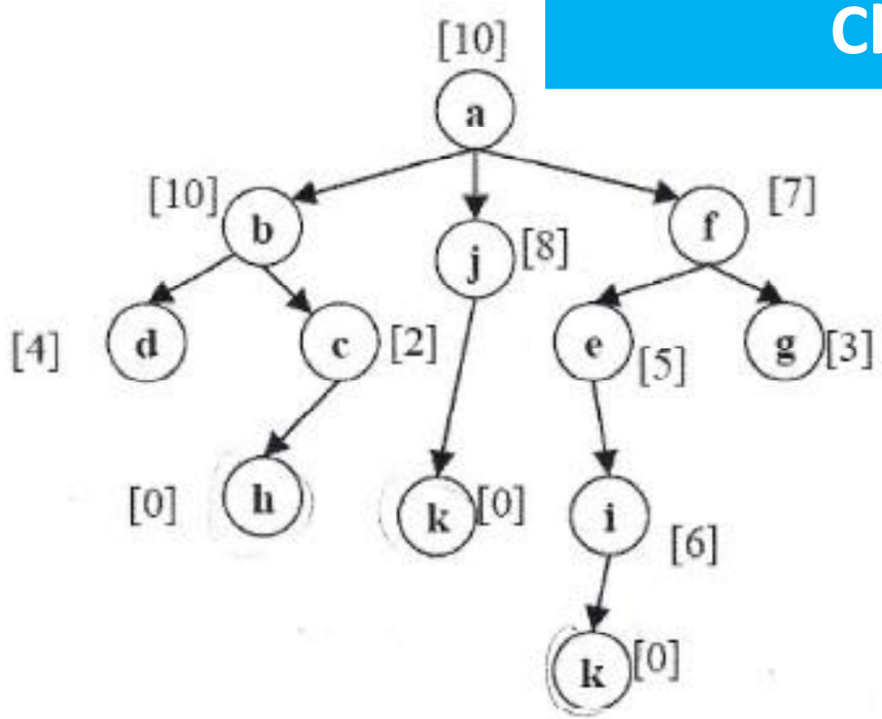


Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

Simulated Annealing

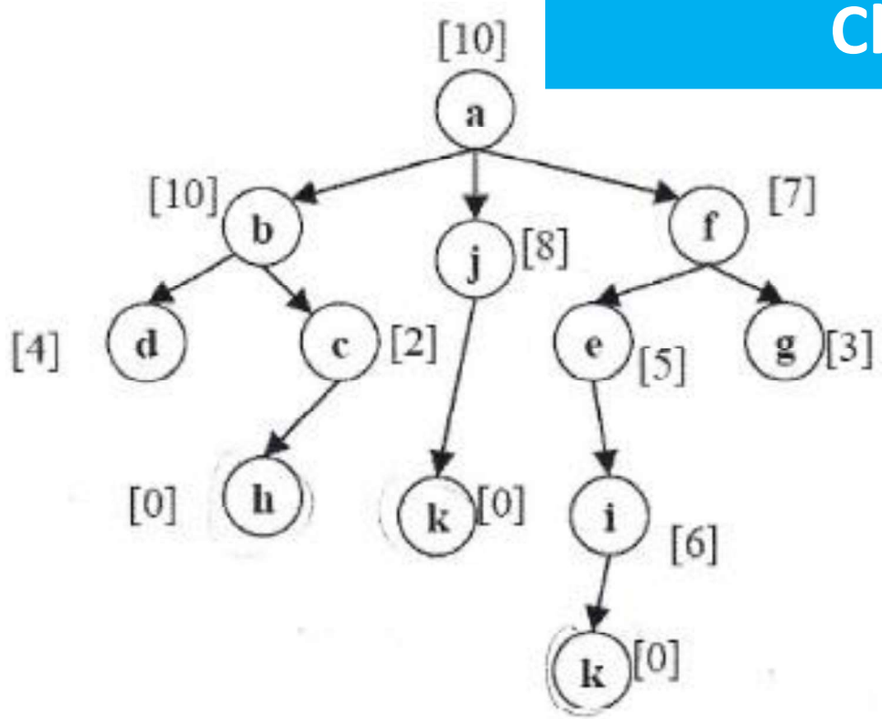
Randomly Select a Child

Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3



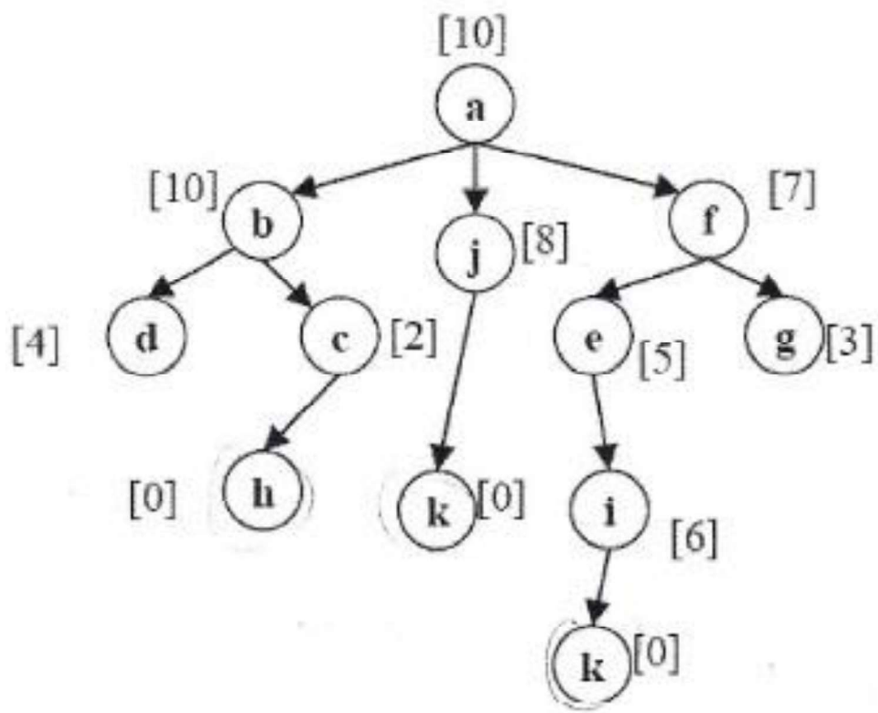
Simulated Annealing

Randomly Select a Child



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

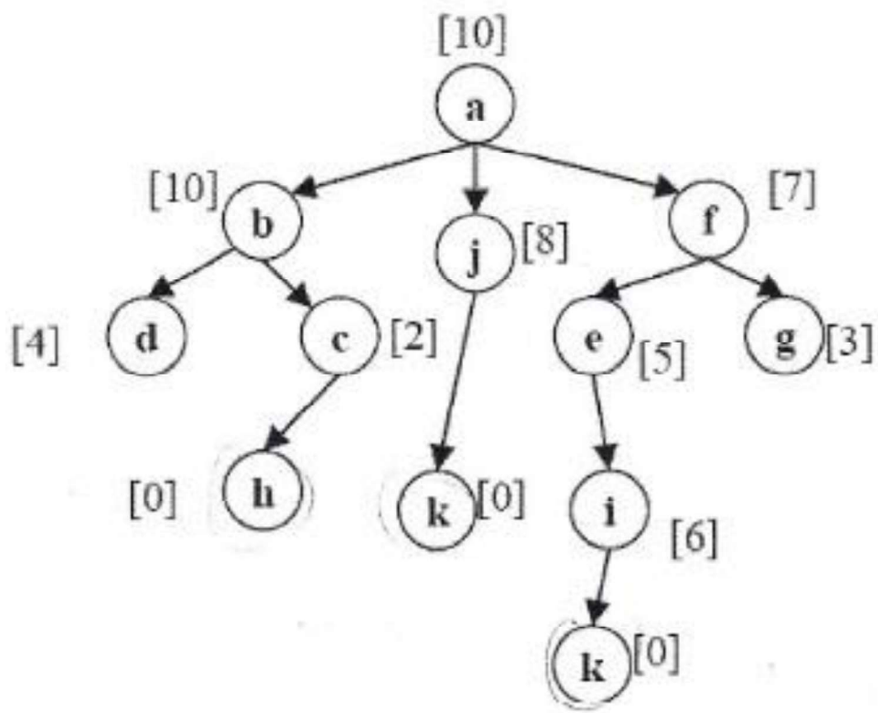
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

Check if next node e_5 is better than current node

Simulated Annealing

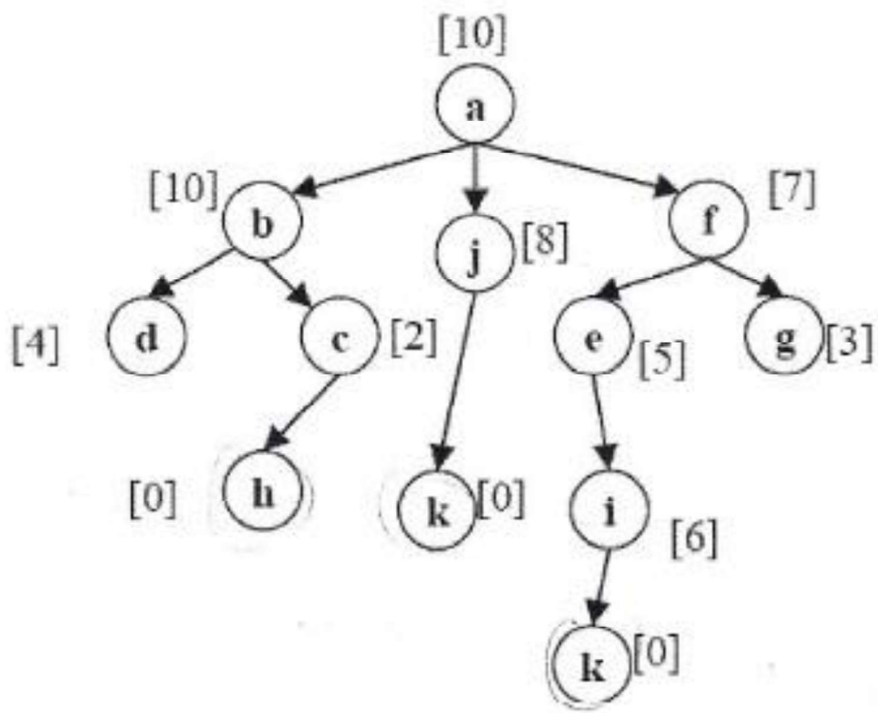


Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

Check if next node e_5 is better than current node

$$\Delta E > 0$$

Simulated Annealing

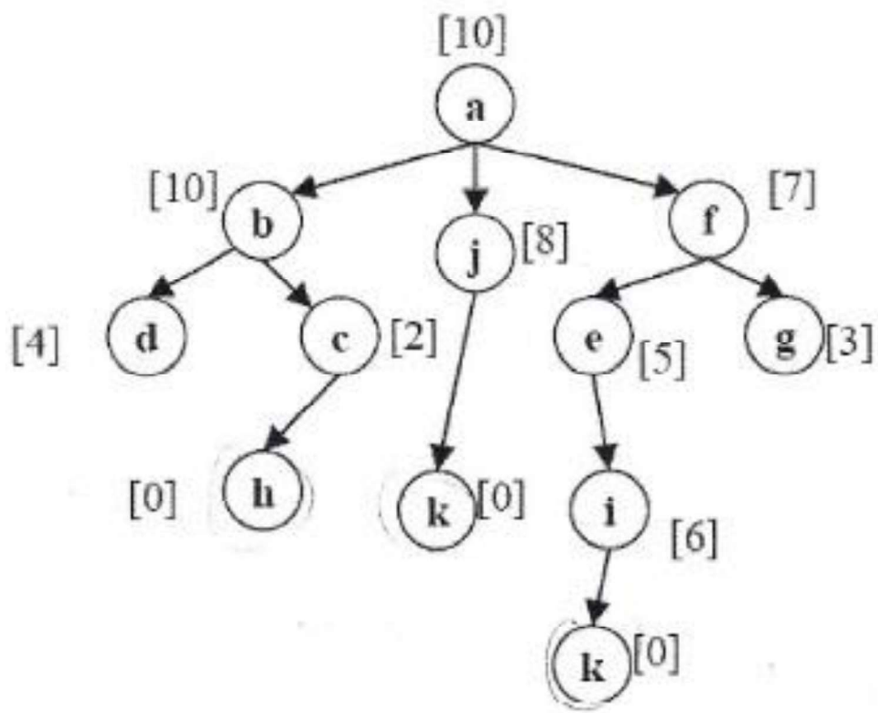


Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

Check if next node e_5 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

Simulated Annealing



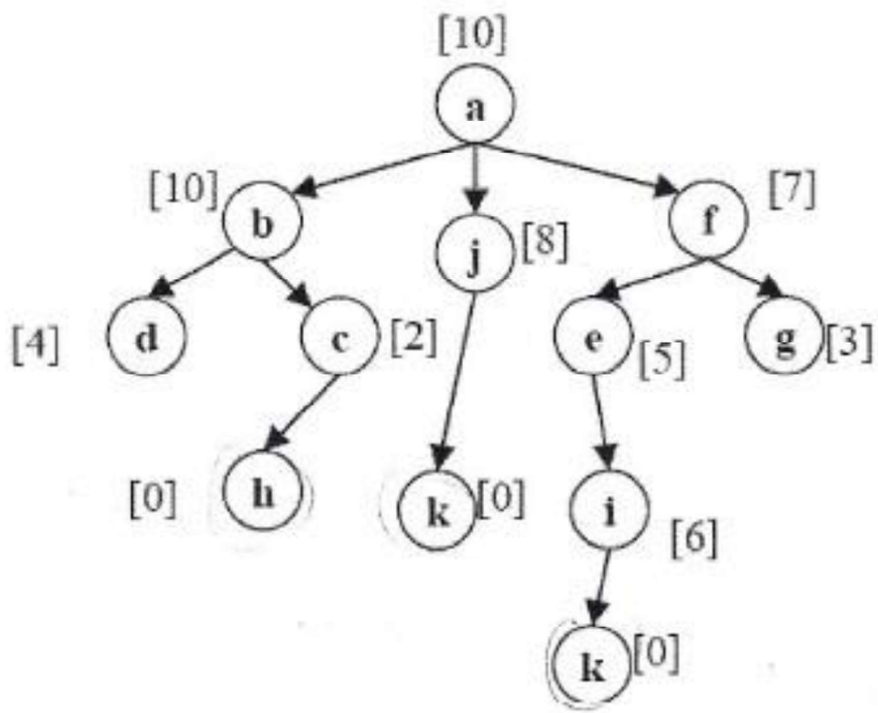
Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

Check if next node e_5 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(e_5) - \text{value}(f_7)$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

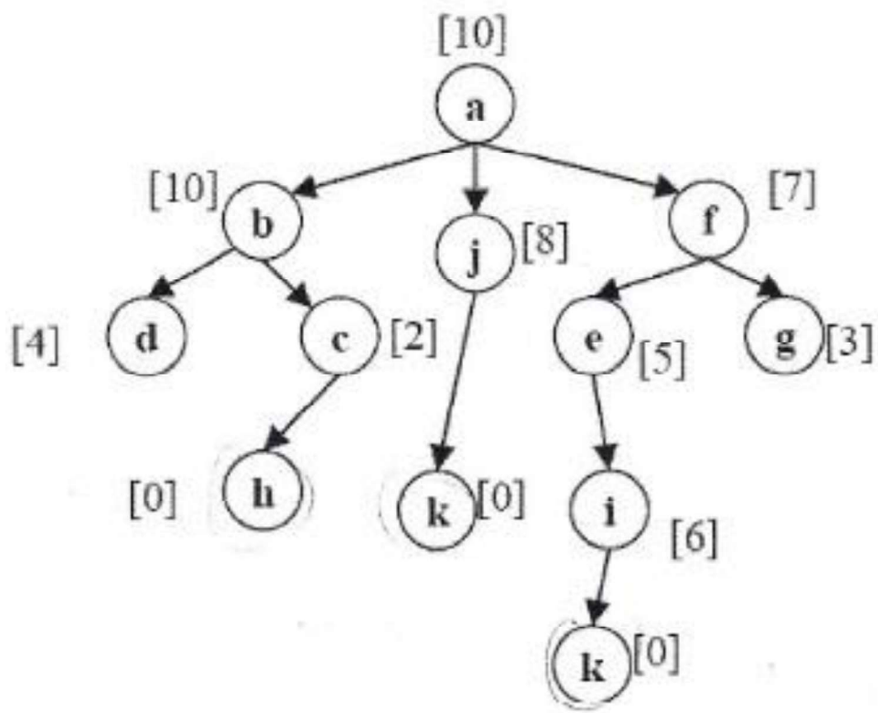
Check if next node e_5 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(e_5) - \text{value}(f_7)$$

$$\text{value}(e_5) = -\text{heuristic}(e_5) = -5$$

Simulated Annealing



$\mathbf{value}(f_7) = -\mathbf{heuristic}(f_7) = -7$

Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

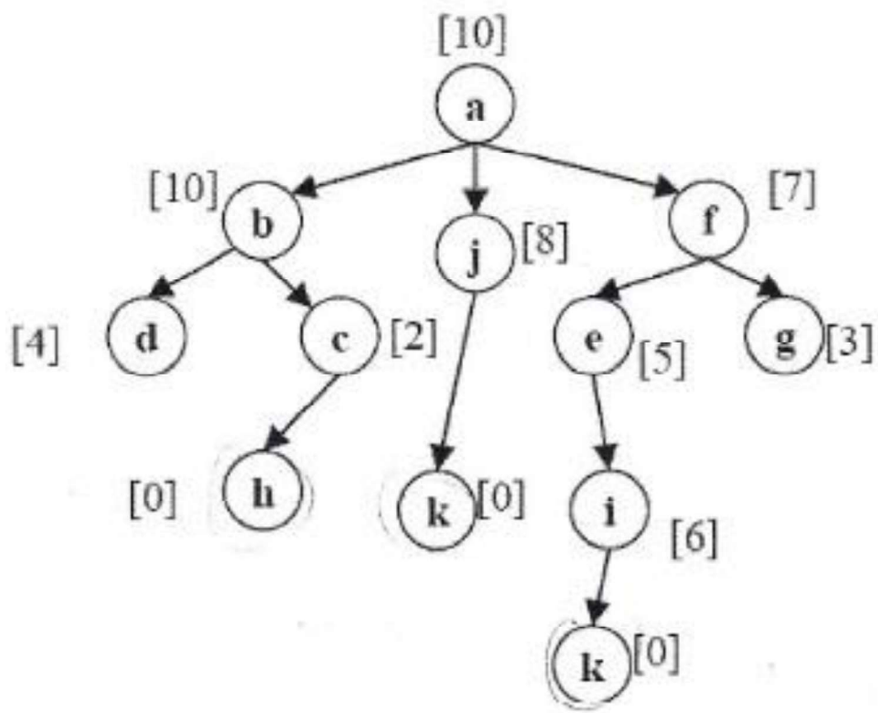
Check if next node e_5 is better than current node

$$\Delta \mathrm{E} = \mathbf{value}(\mathbf{next}) - \mathbf{value}(\mathbf{current})$$

$$\Delta \mathrm{E} = \mathbf{value}(e_5) - \mathbf{value}(f_7)$$

$$\mathbf{value}(e_5) = -\mathbf{heuristic}(e_5) = -5$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

Check if next node e_5 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

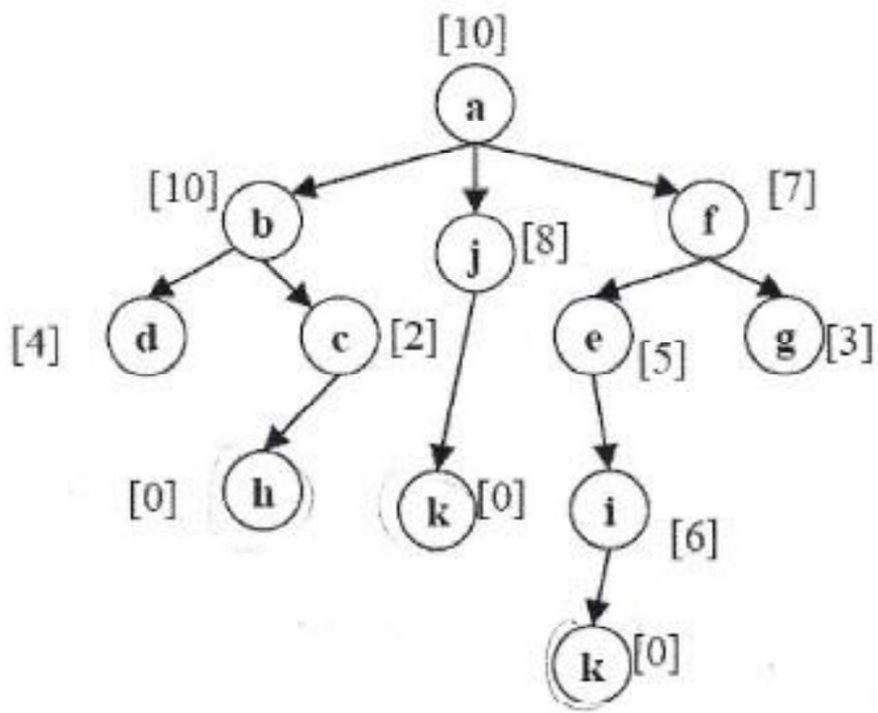
$$\Delta E = \text{value}(e_5) - \text{value}(f_7)$$

$$\Delta E = -5 - (-7) = +2$$

Simulated Annealing

$\because \Delta E > 0$

$\therefore$ e_5 will be selected with probability 1



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3

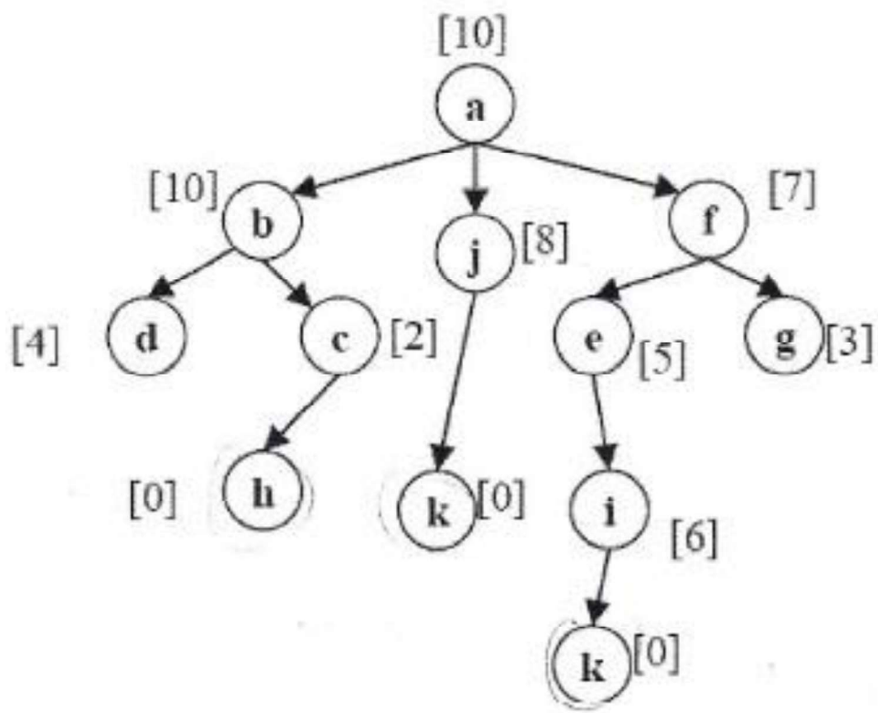
Check if next node e_5 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(e_5) - \text{value}(f_7)$$

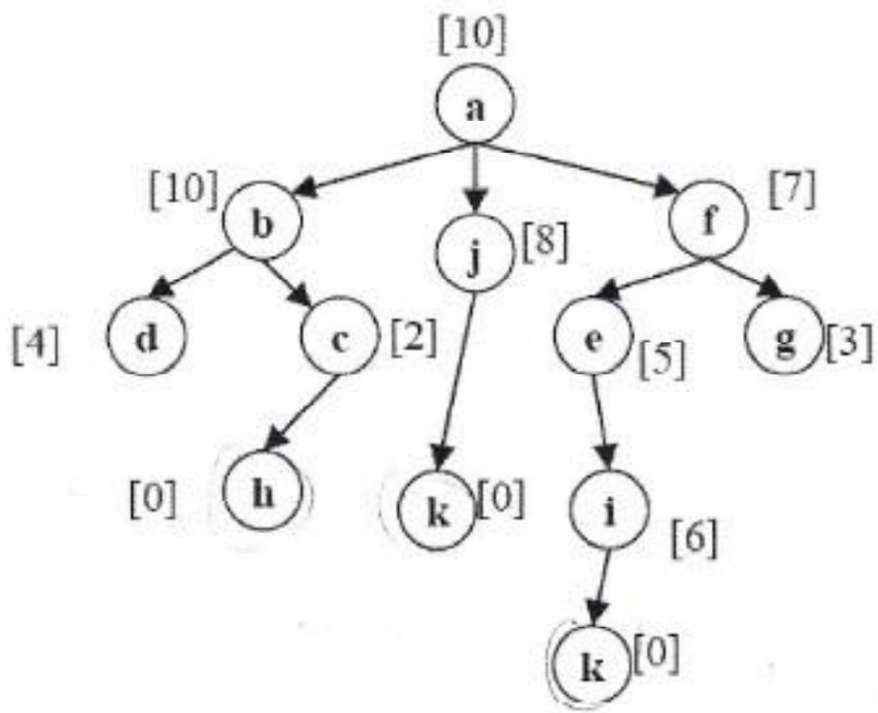
$$\Delta E = -5 - (-7) = +2$$

Simulated Annealing



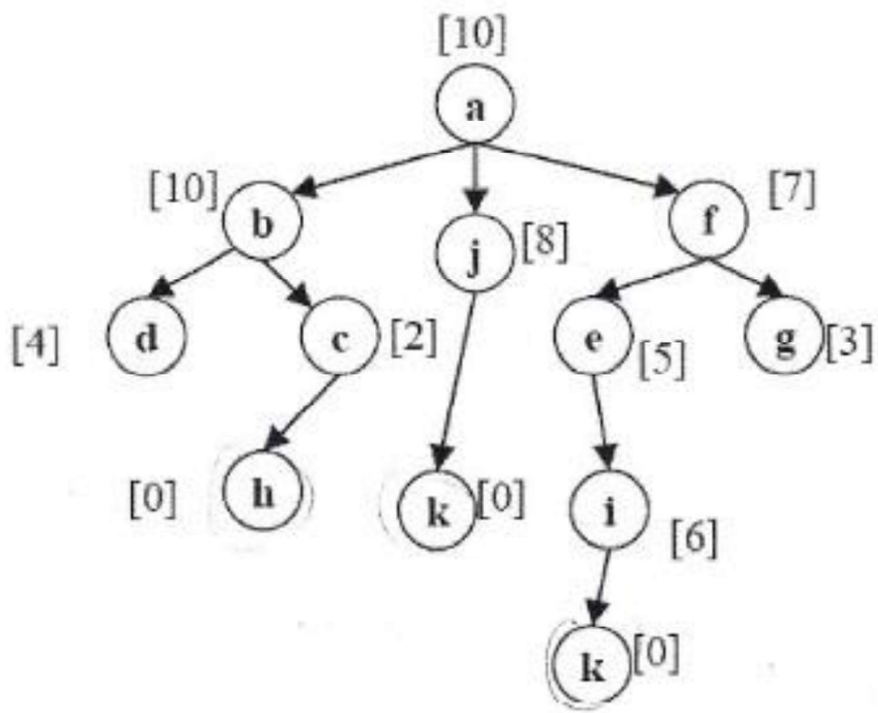
Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	

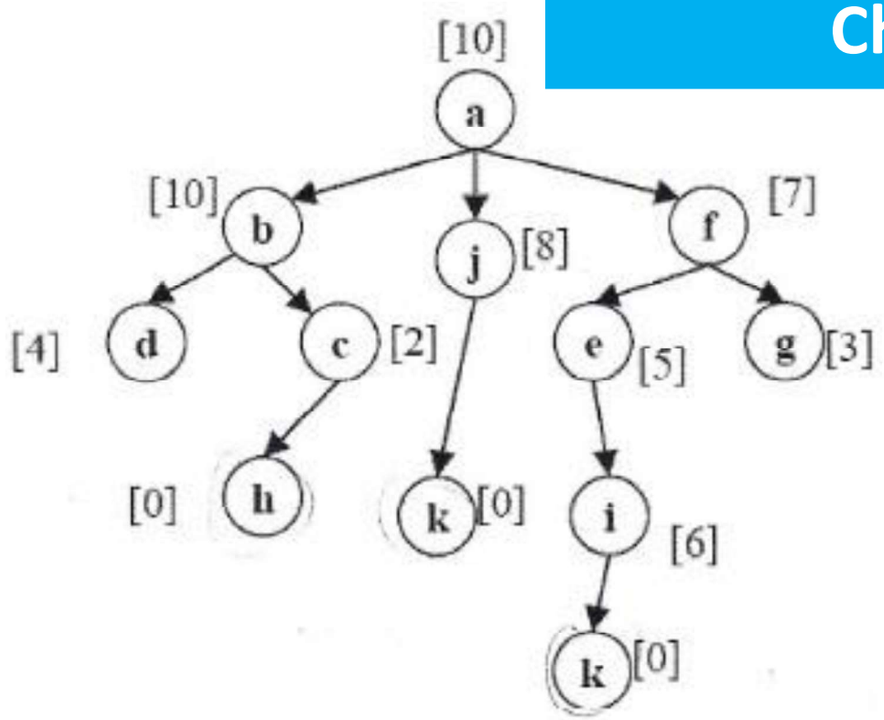
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Simulated Annealing

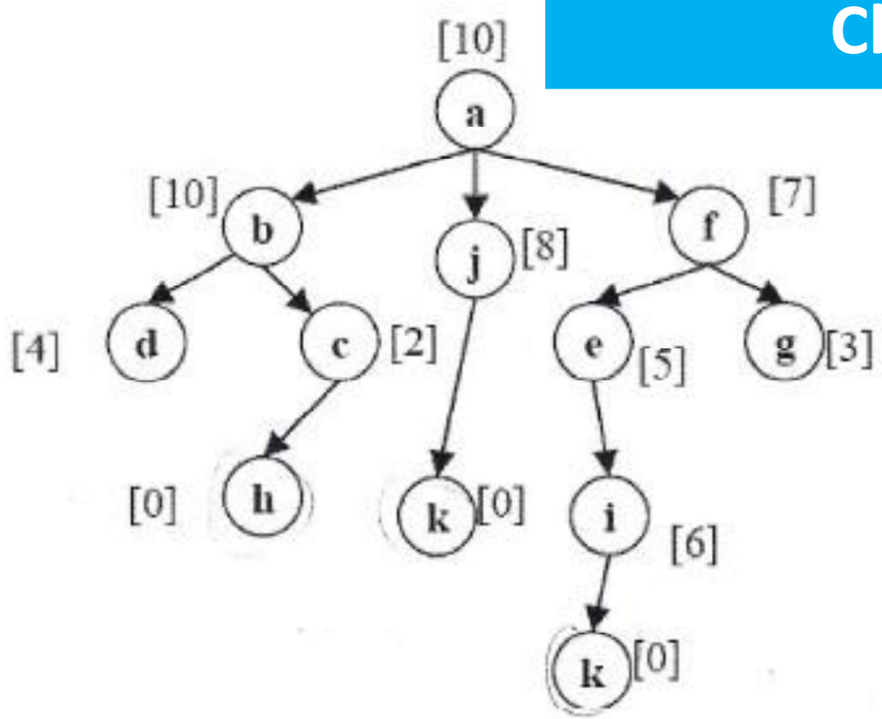
Randomly Select a Child



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

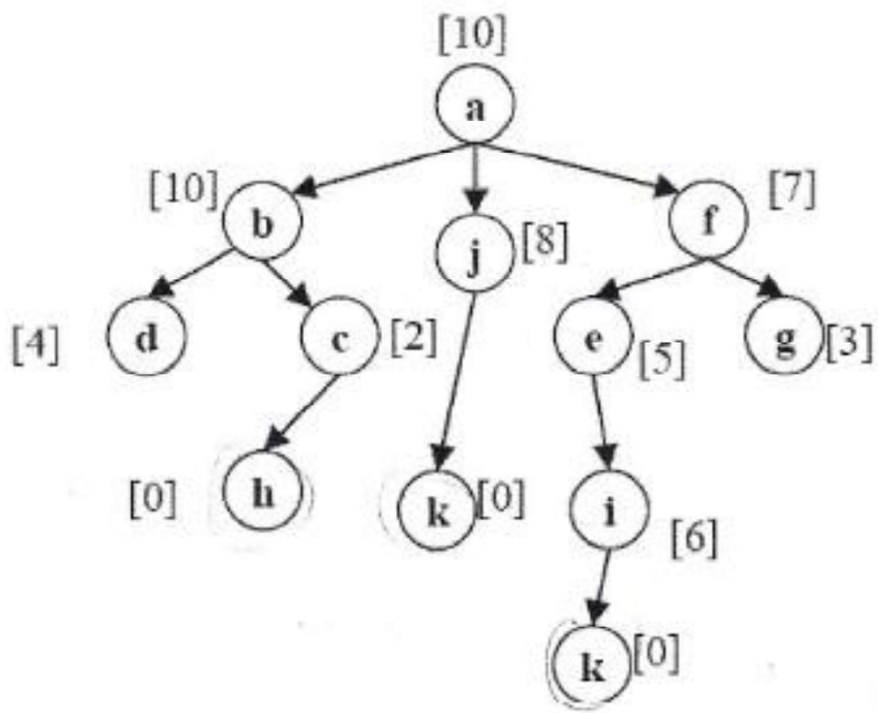
Simulated Annealing

Randomly Select a Child



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

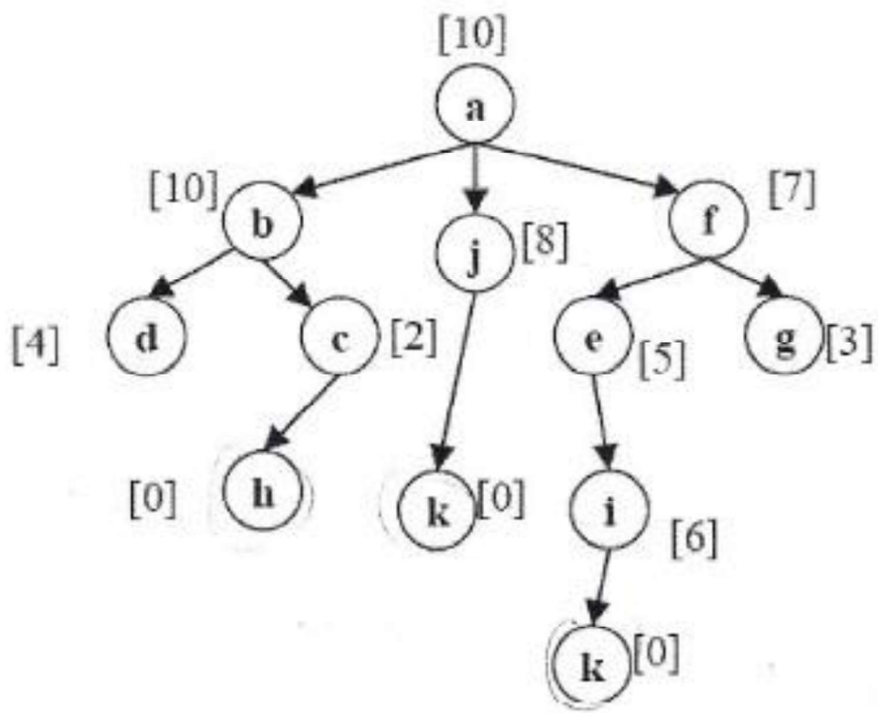
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node i_6 is better than current node

Simulated Annealing

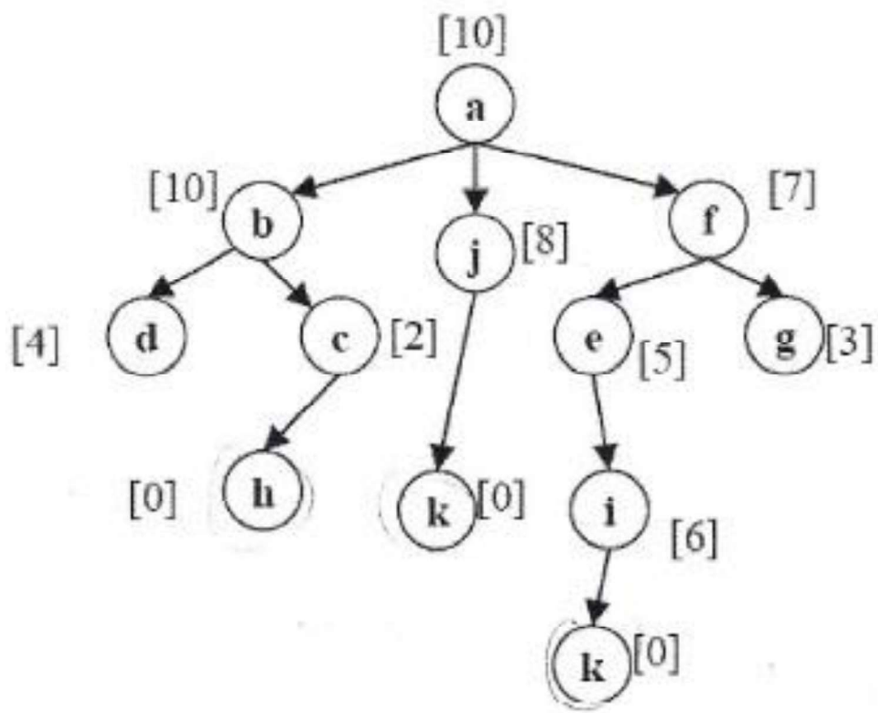


Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node i_6 is better than current node

$$\Delta E > 0$$

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node i_6 is better than current node

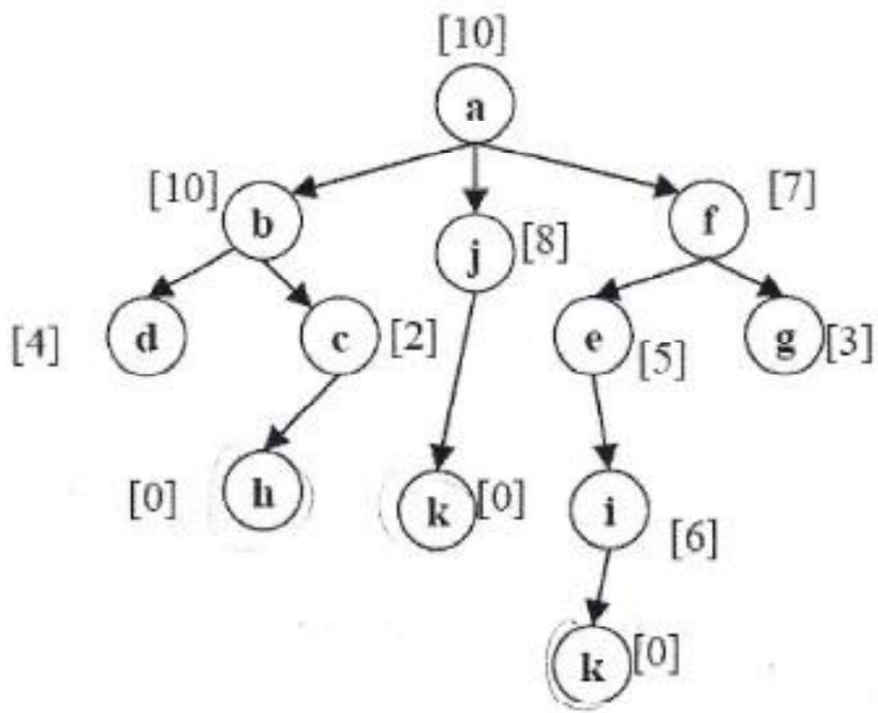
$$\Delta E > 0$$

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(i_6) - \text{value}(e_5)$$

Simulated Annealing

$$\text{value}(i_6) = -\text{heuristic}(i_6) = -6$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node i_6 is better than current node

$$\Delta E > 0$$

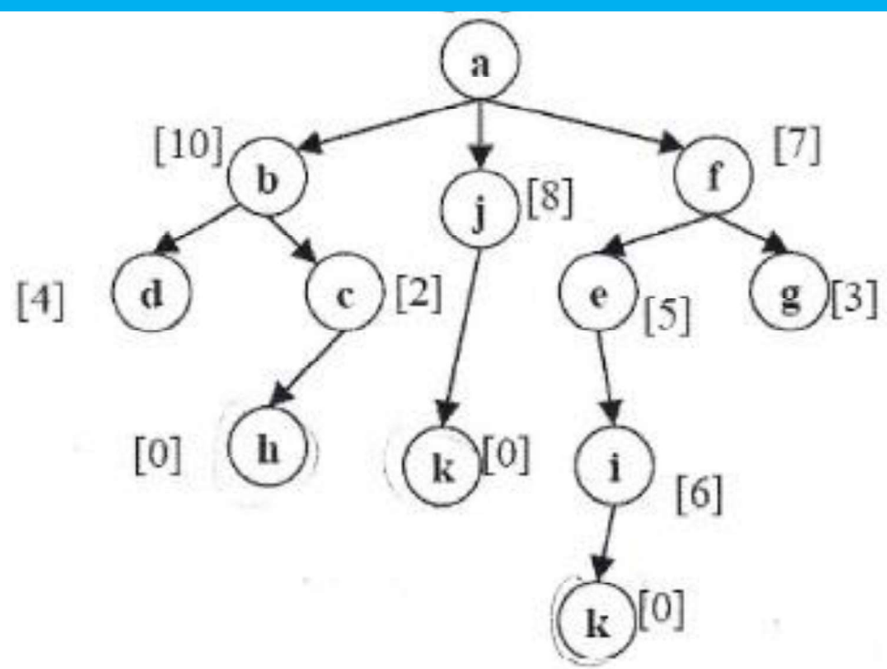
$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(i_6) - \text{value}(e_5)$$

Simulated Annealing

$$\text{value}(i_6) = -\text{heuristic}(i_6) = -6$$

$$\text{value}(e_5) = -\text{heuristic}(e_5) = -5$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node i_6 is better than current node

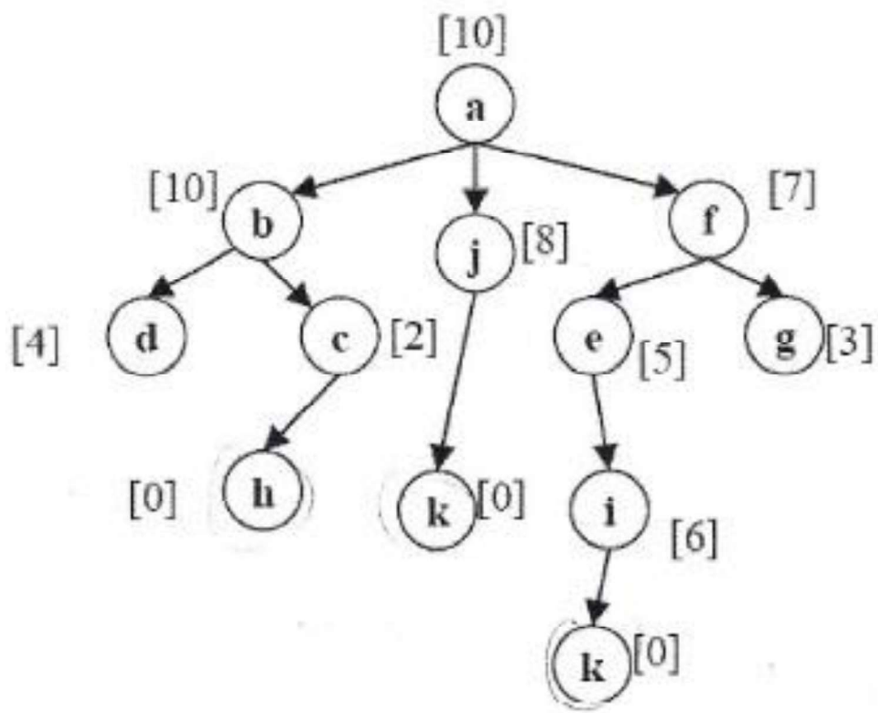
$$\Delta E > 0$$

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(i_6) - \text{value}(e_5)$$

Simulated Annealing

$$\Delta E = -6 - (-5) = -1$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node i_6 is better than current node

$$\Delta E > 0$$

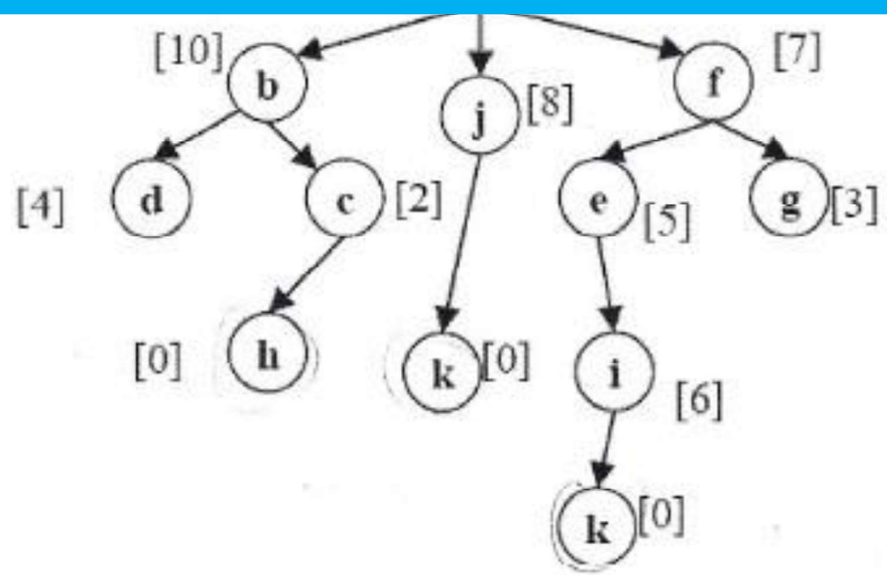
$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(i_6) - \text{value}(e_5)$$

Simulated Annealing

$\because \Delta E < 0$

$\therefore i_6$ can be selected with probability $p = e^{\frac{\Delta E}{T}}$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node e_5 is better than current node

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

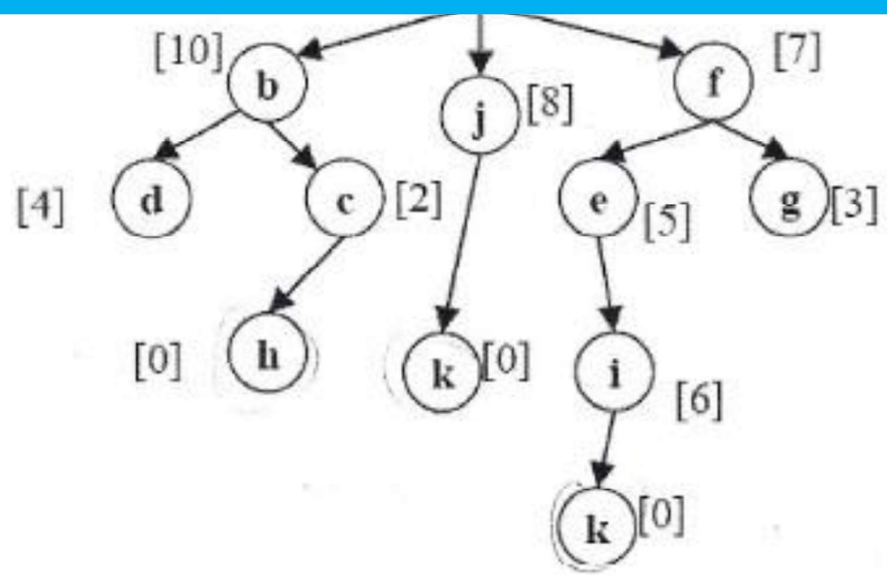
$$\Delta E = \text{value}(e_5) - \text{value}(f_7)$$

$$\Delta E = -5 - (-7) = +2$$

Simulated Annealing

$\because \Delta E < 0$

$\therefore$ i_6 can be selected with probability $p = e^{\frac{\Delta E}{T}}$

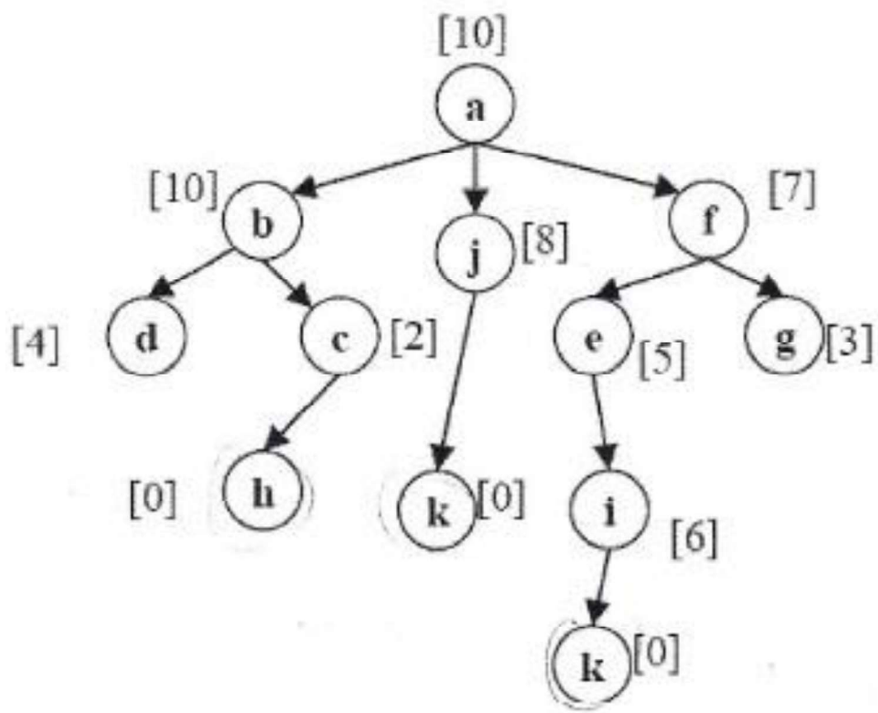


Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

Check if next node e_5 is better than current node

$$p = e^{\frac{-1}{10}} = e^{\frac{-1}{10}} = .905$$

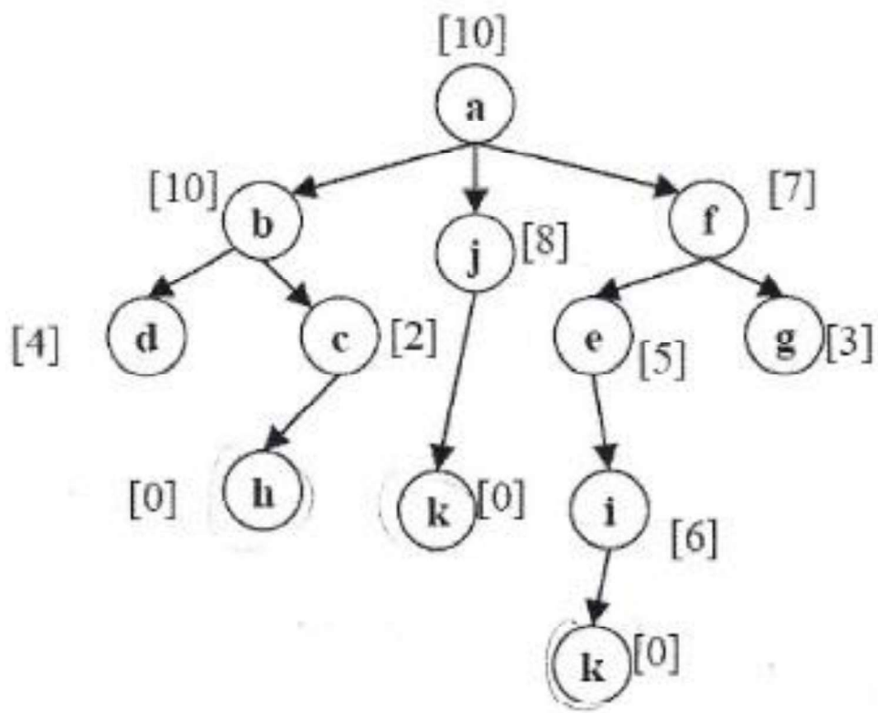
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6

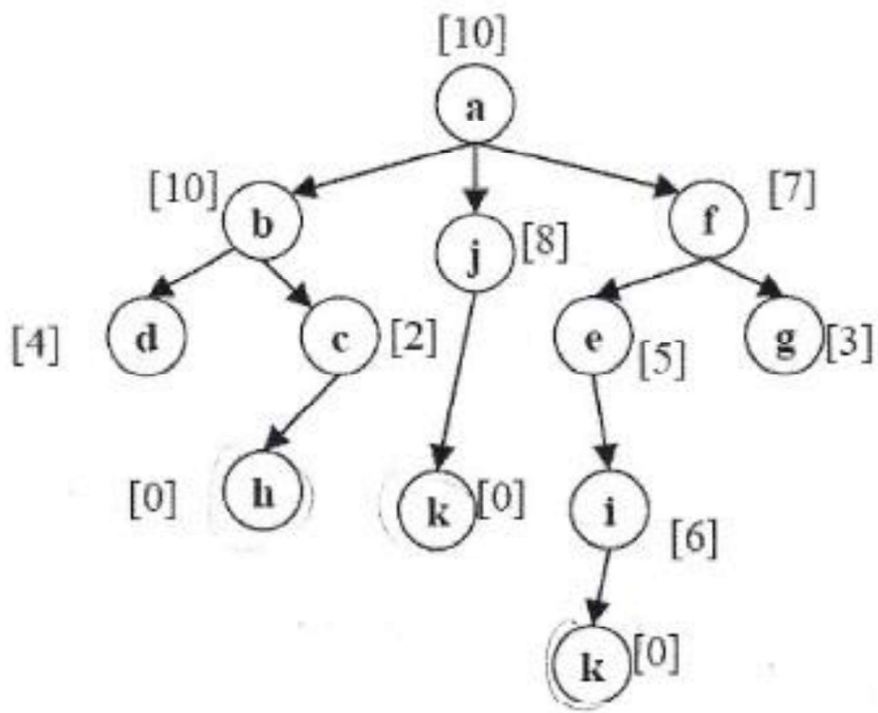
Because the only child of e_5 is i_6 then it will be selected even if its probability is not 1.

Simulated Annealing



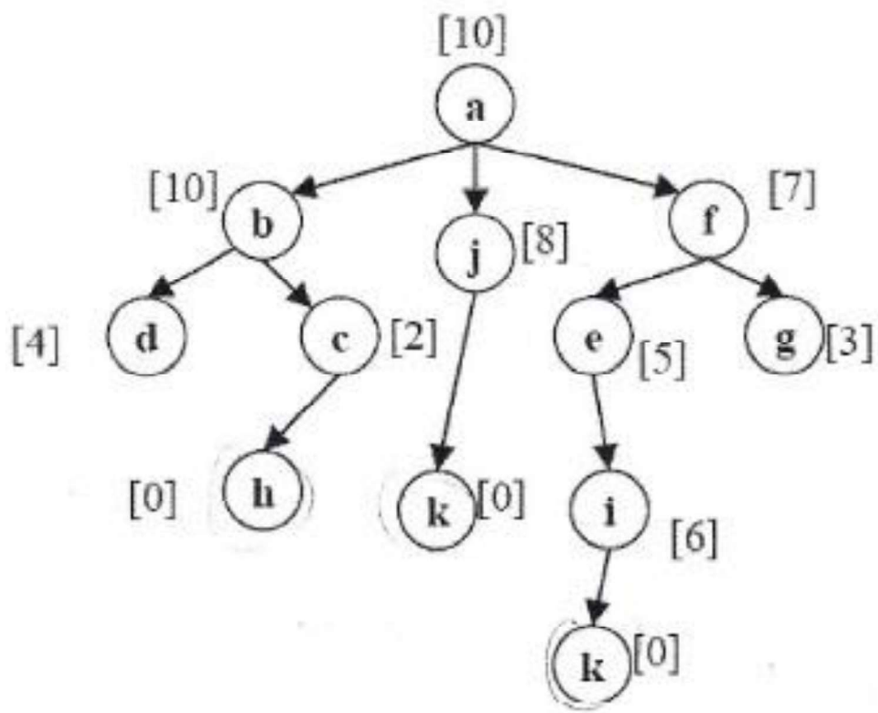
Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	

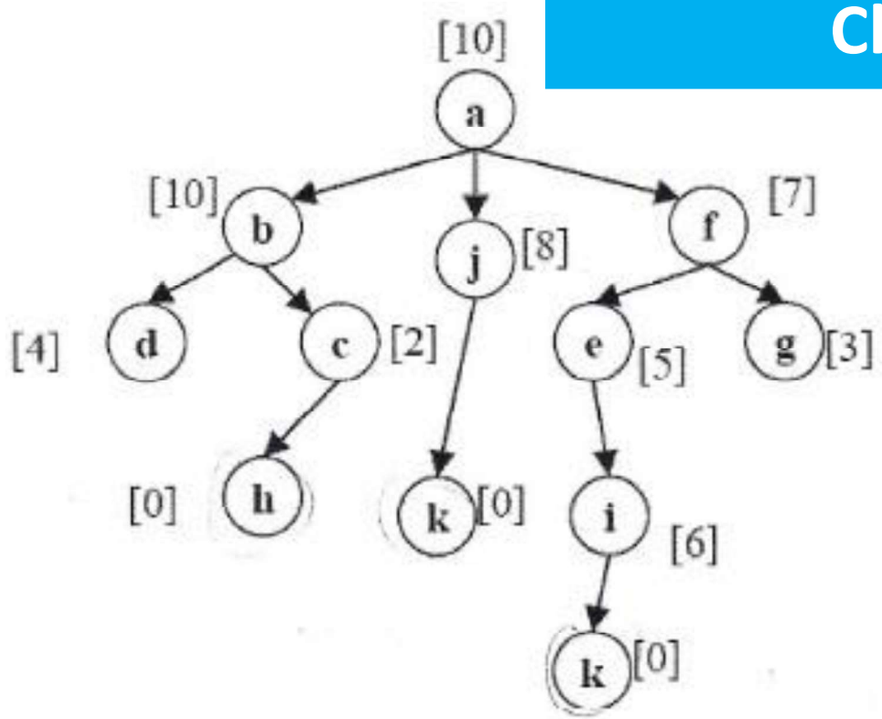
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Simulated Annealing

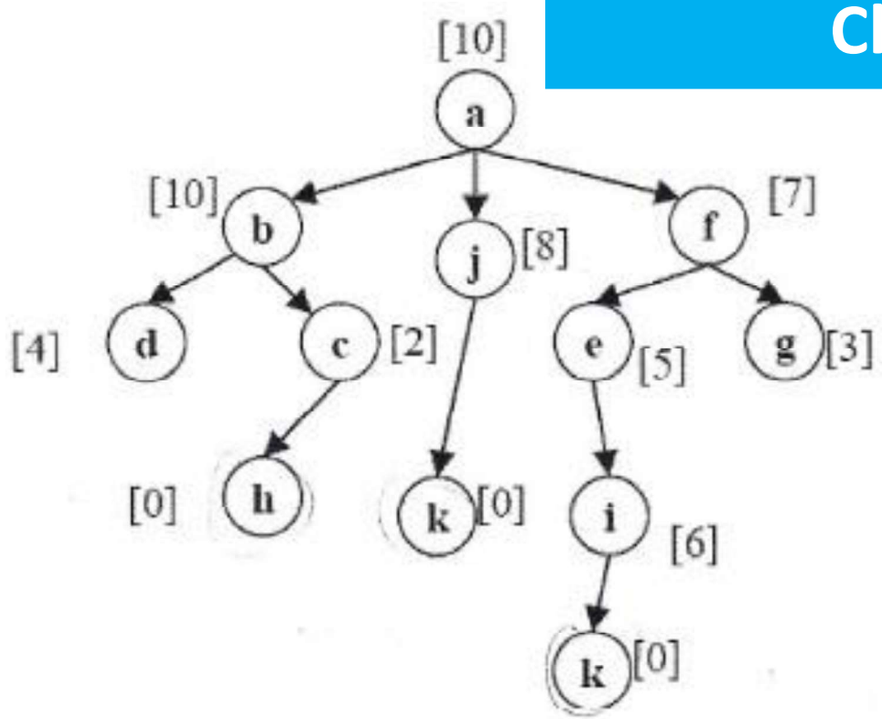
Randomly Select a Child



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

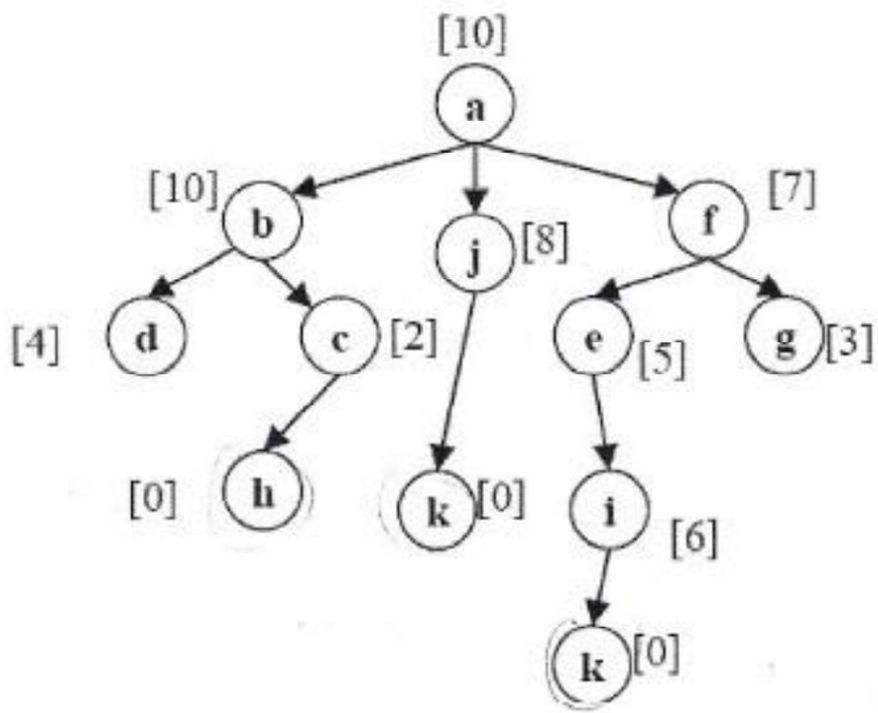
Simulated Annealing

Randomly Select a Child



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

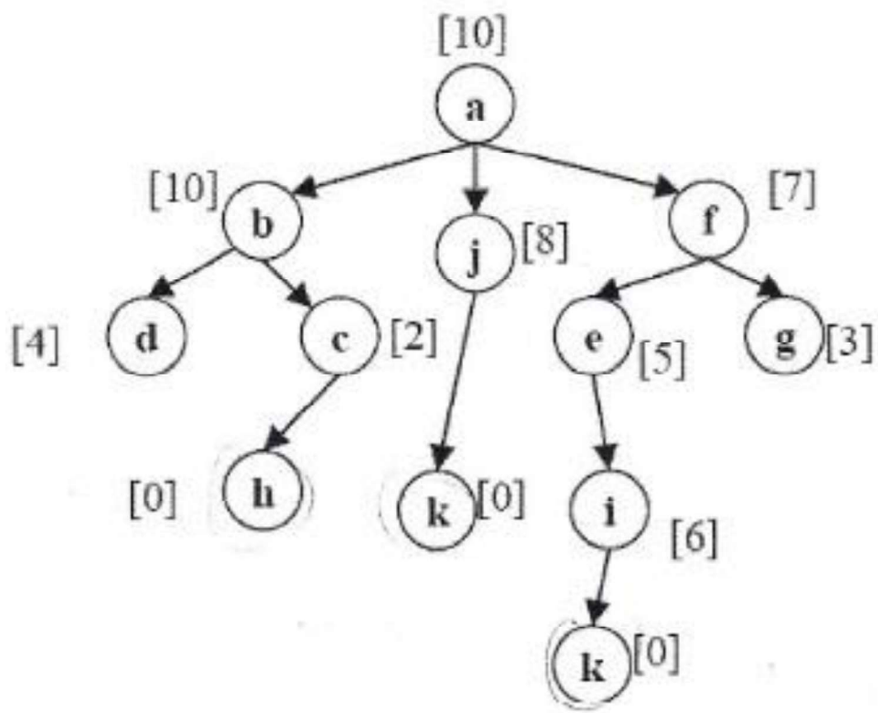
Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

Simulated Annealing



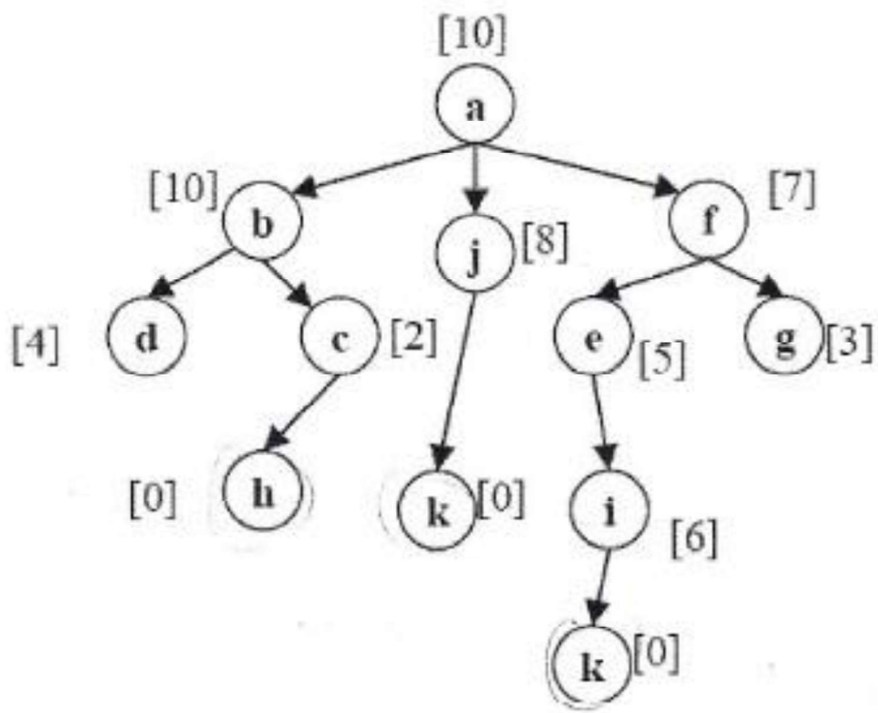
Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

$$\Delta E > 0$$

Simulated Annealing

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

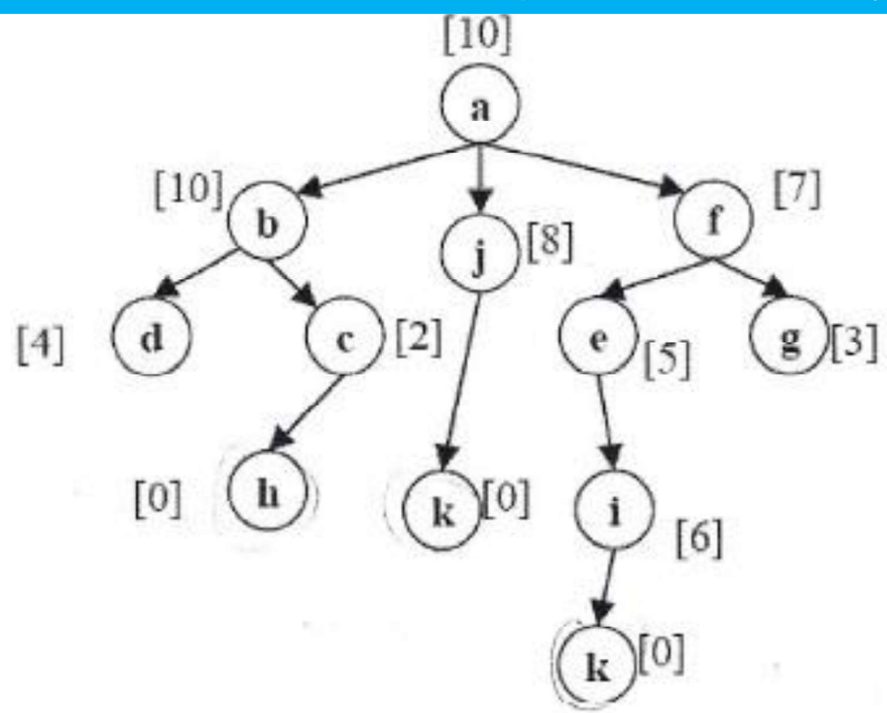
Check if next node k_0 is better than current node

$$\Delta E > 0$$

Simulated Annealing

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(k_0) - \text{value}(i_6)$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

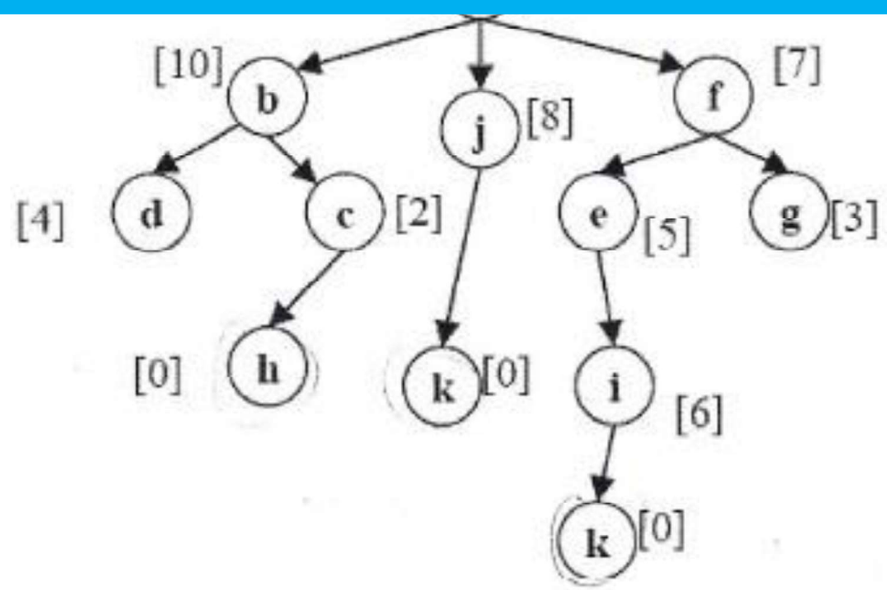
$$\Delta E > 0$$

Simulated Annealing

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(k_0) - \text{value}(i_6)$$

$$\text{value}(k_0) = -\text{heuristic}(k_0) = 0$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

$$\Delta E > 0$$

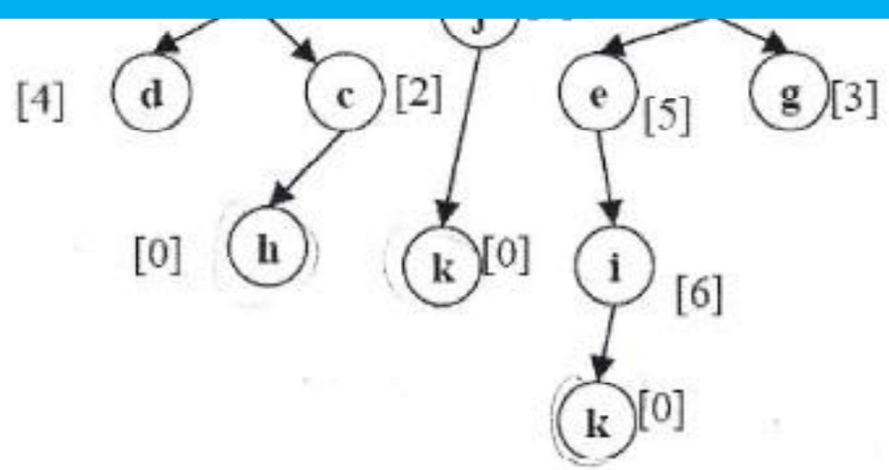
Simulated Annealing

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(k_0) - \text{value}(i_6)$$

$$\text{value}(k_0) = -\text{heuristic}(k_0) = 0$$

$$\text{value}(i_6) = -\text{heuristic}(i_6) = -6$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

$$\Delta E > 0$$

Simulated Annealing

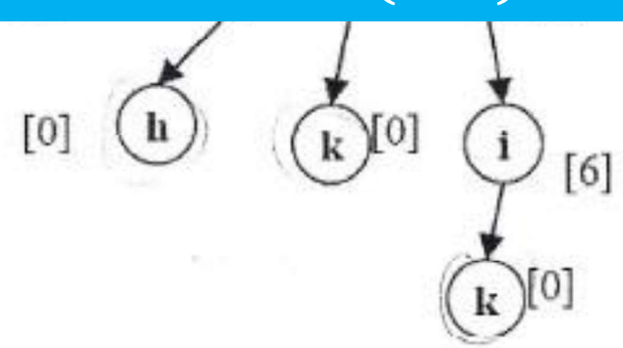
$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(k_0) - \text{value}(i_6)$$

$$\text{value}(k_0) = -\text{heuristic}(k_0) = 0$$

$$\text{value}(i_6) = -\text{heuristic}(i_6) = -6$$

$$\Delta E = 0 - (-6) = +6$$



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

$$\Delta E > 0$$

Simulated Annealing

$$\Delta E = \text{value}(\text{next}) - \text{value}(\text{current})$$

$$\Delta E = \text{value}(k_0) - \text{value}(i_6)$$

$$\text{value}(k_0) = -\text{heuristic}(k_0) = 0$$

$$\text{value}(i_6) = -\text{heuristic}(i_6) = -6$$

$$\Delta E = 0 - (-6) = +6$$

$\because \Delta E > 0$

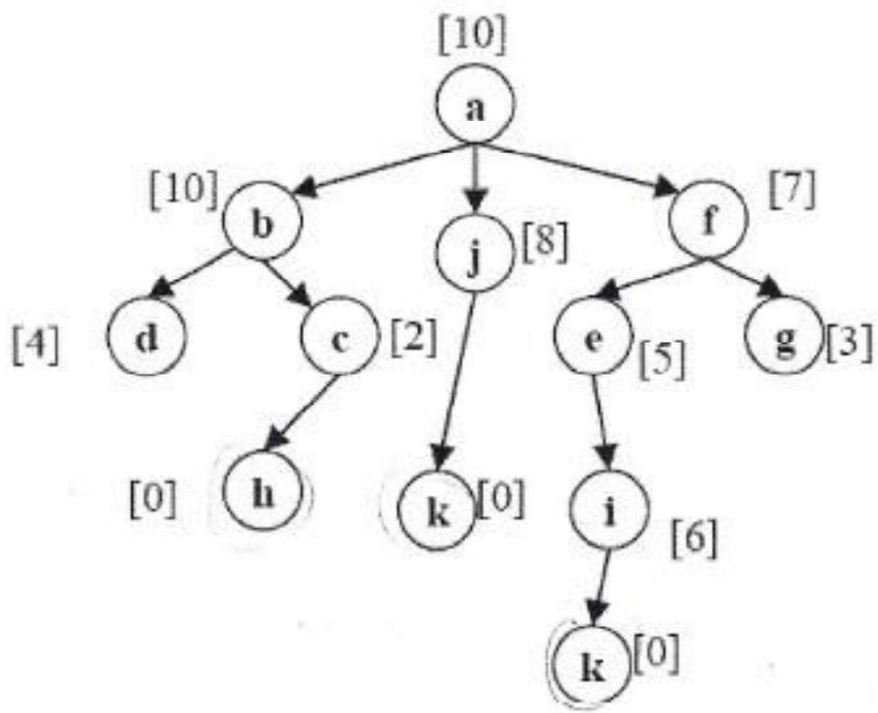
$\therefore$ k_0 will be selected with probability 1

Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0

Check if next node k_0 is better than current node

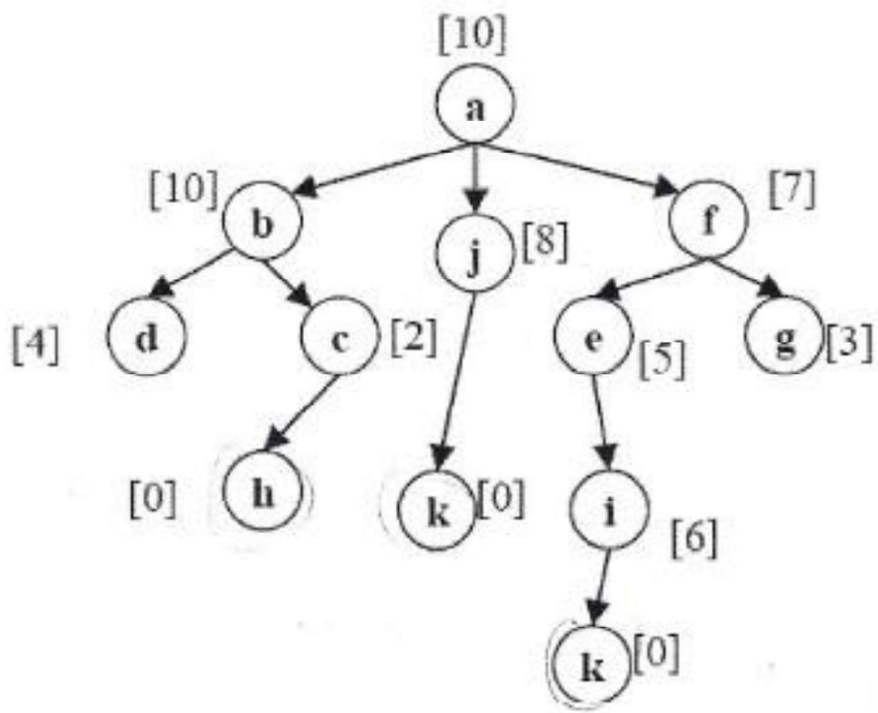
$$\Delta E > 0$$

Simulated Annealing



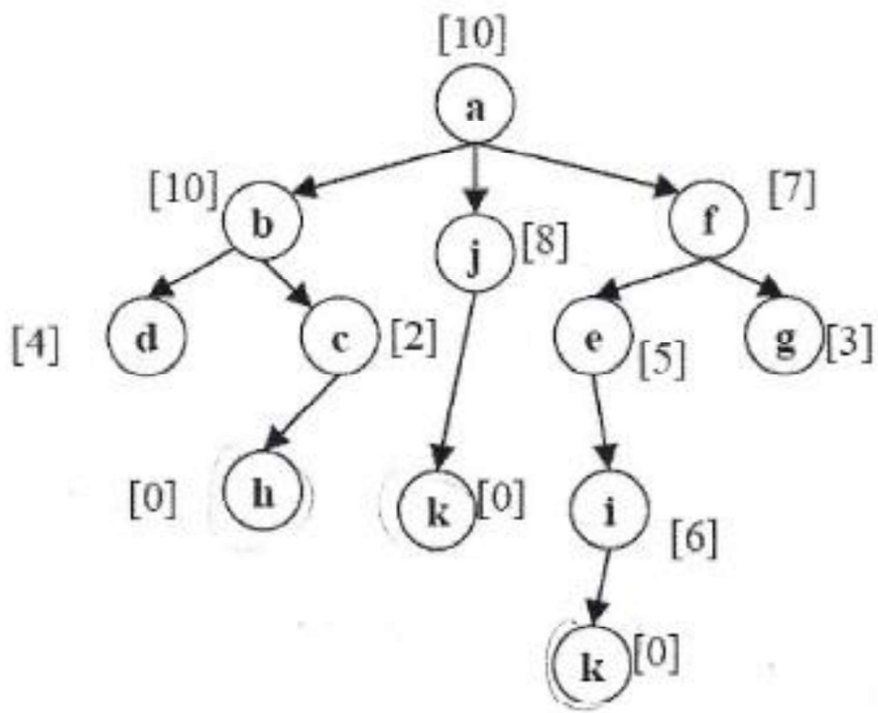
Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0
k	

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0
k	

Simulated Annealing



Current	Children
a	---
a	f_7, j_8, b_{10}
f	e_5, g_3
e	i_6
i	k_0
k	

GOAL